



BANDARI MARITIME ACADEMY

COMPETENCY-BASED CURRICULUM

FOR

NAUTICAL SCIENCE

KNQF LEVEL 5

PROGRAMME CODE - 1041 454 A

EXCELLENCE IN MARITIME TRAINING & DEVELOPMENT

First published 2024

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FOREWORD

It is with great pleasure and a sense of responsibility that I introduce the Competency-Based Curriculum (CBC) for the Bandari Maritime Academy. In an era where our global economy is increasingly reliant on maritime trade and transportation, the significance of a competent workforce in the maritime sector cannot be overstated. This curriculum represents a vital step forward in ensuring that our workforce is equipped with the necessary skills and knowledge to navigate the complexities of the maritime industry effectively.

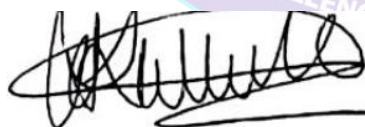
The development of this CBC is a testament to our commitment to fostering a highly skilled workforce that can meet the challenges and demands of the maritime sector in the 21st century.

One of the key features of this curriculum is its competency-based approach. By focusing on the specific skills and abilities required for success in various roles within the maritime industry, this curriculum ensures that learning is practical, relevant, and aligned with industry standards. Whether it is in ship operations, port management, maritime law, or marine engineering, the CBC provides learners with a comprehensive framework for acquiring the competencies needed to excel in their chosen fields.

Moreover, this curriculum is designed to be flexible and adaptable to the evolving needs of the maritime industry. As technology advances and new challenges emerge, it is essential that our workforce remains agile and capable of responding to change. The CBC facilitates lifelong learning and professional development, enabling individuals to stay abreast of industry trends and advancements throughout their careers.

I would like to express my sincere appreciation to all those who contributed to the development of this curriculum, including educators, industry professionals, policymakers, and stakeholders. Your expertise, insights, and dedication have been instrumental in shaping this curriculum into a robust and forward-thinking framework for ship operations and management.

In conclusion, I am confident that the Competency-Based Curriculum for the Bandari Maritime Academy will serve as a cornerstone for excellence in maritime education and training. By equipping our workforce with the requisite skills and competencies, we not only enhance individual opportunities but also strengthen the overall competitiveness and resilience of our maritime industry.



PRINCIPAL SECRETARY
STATE DEPARTMENT FOR SHIPPING AND MARITIME AFFAIRS
MINISTRY OF MINING, BLUE ECONOMY AND MARITIME AFFAIRS

PREFACE

Kenya Vision 2030 aims to transform the country into a newly industrializing, “middle-income country providing a high-quality life to all its citizens by the year 2030”. Kenya intends to create a globally competitive and adaptive human resource base to meet the requirements of a rapidly industrializing economy through life-long education and training. TVET has a responsibility to facilitate the process of inculcating knowledge, skills, and attitudes necessary for catapulting the nation to a globally competitive country, hence the paradigm shift to embrace Competency-Based Education and Training (CBET).

The Technical and Vocational Education and Training Act No. 29 of 2013 and Sectional Paper No. 1 of 2019 on Reforming Education and Training in Kenya, emphasized the need to reform curriculum development, assessment, and certification. This called for a shift to CBET in order to address the mismatch between skills acquired through training and skills needed by industry as well as increase the global competitiveness of the Kenyan labour force.

Bandari Maritime Academy in conjunction with, Kenya Maritime Authority, the Nautical Science Industrial Advisory Committee (IAC), Industry representatives, and trainers have developed this curriculum.

This curriculum is designed and organized with an outline of learning outcomes; suggested delivery methods, training/learning resources and methods of assessing the trainee’s achievement. The curriculum is competency-based and allows multiple entry and exit to the course.

I am grateful to the Board members, BMA staff, Kenya Maritime Authority, Nautical Science IAC, trainers, expert workers and all those who participated in the development of this curriculum.



BOARDS CHAIRPERSON
BANDARI MARITIME ACADEMY

ACKNOWLEDGEMENT

As the Chief Executive Officer of Bandari Maritime Academy, I am honoured to extend my heartfelt appreciation and acknowledgment for the development of the Competency-Based Curriculum (CBC).

The introduction of this curriculum marks a significant milestone in the advancement of maritime transport operations training in our nation. It reflects a collaborative effort among educators, industry experts, policymakers, and stakeholders to ensure that our maritime workforce is equipped with the requisite skills and knowledge to thrive in a rapidly evolving industry landscape.

At Bandari Maritime Academy, we are committed to excellence in maritime transport and logistics, and the CBC aligns perfectly with our mission to produce highly skilled professionals who can contribute effectively to the maritime sector. By emphasizing competency-based learning, the curriculum provides our students with a clear roadmap for acquiring the practical skills and expertise needed to excel in their chosen careers.

I would like to express my gratitude to all those who played a role in the development of this curriculum, including Kenya Maritime Authority, Nautical Science Industrial Advisory Committee (IAC) members, industry representatives, and trainers for their contribution to the development of this curriculum. Your collective efforts have resulted in a curriculum that is not only comprehensive and relevant but also adaptable to the evolving needs of the maritime industry.

As we embark on the implementation of this curriculum at Bandari Maritime Academy, I am confident that it will serve as a catalyst for innovation and excellence in Nautical Science. By empowering our students with the competencies demanded by the industry, we are not only preparing them for success but also contributing to the growth and prosperity of the maritime sector as a whole.

I extend my best wishes for the successful implementation of the Competency-Based Curriculum in the Academy. Together, let us continue to raise the standards of excellence in Nautical Science, by ensuring a bright and promising future for the maritime industry.



**DIRECTOR GENERAL/CEO
BANDARI MARITIME ACADEMY**

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COURSE OVERVIEW

The Able Seafarer Deck Rating qualification KNQF level 5 consists of competencies that a person must achieve to enable him/her to be certified as a Deck Rating. This qualification pack entails performing seamanship Practices, performing safe watchkeeping practices, performing the safe operation and maintenance of deck machinery, cargo handling gear and equipment, performing berthing, anchoring and other mooring operations, performing cargo handling and stowage, performing ship handling and maneuvering, responding to emergencies at sea and in port and performing Basic chartwork Practices.

The course consists of basic, common and core units of learning as indicated below:

Basic Units of Learning

Unit code	Unit of learning title	Notional hours	Credits
0031 441 01A	Communication Skills	30	3
0611 441 02A	Information, Technology and Communication Skills	50	5
0413 441 03A	Entrepreneurship Skills	50	7
0031 441 04A	Employability Skills	50	5
1022 441 05A	Occupational Safety and Health Practices	30	3
0712 451 06A	Marine environment Protection Principles	50	5
Total		260	26

Common Units of Learning

Unit Code	Unit of learning title	Duration in Hours	Credits
1041 451 07A	Maritime mathematics	80	8
1041 451 08A	Marine electrical and electronics principles	70	7
1041 451 09A	Ship Construction principles	70	7
1041 451 10A	Ship stability	80	8
1041 451 11A	Basic sea safety (STCW)	50	5
1041 451 12A	Meteorological principles	60	6
1041 451 13A	Maritime Communication Skills	40	4
Total		450	45

Core Units of Learning

Core Units of Learning			
1041 451 14A	Seamanship Practices	80	8
1041 451 15A	Watchkeeping practices	80	8
1041 451 16A	Deck machinery, cargo handling gear and equipment	70	7

1041 451 17A	Perform berthing, anchoring and other mooring operations	70	7
1041 451 18A	Perform cargo handling and stowage	80	8
1041 451 19A	Perform ship handling and Maneuvering	70	7
1041 451 20A	Respond to emergencies at sea and in port	60	6
1041 451 21A	Perform Basic chartwork Practices	80	8
TOTAL		590	59
0716 454 18 A	Industrial Attachment	480	48.0
Grand total		1780	178

Course attendance

The trainees MUST attend a minimum of 85% for each of the units of learning to be eligible to sit for the final end of term/semester assessment.

Entry Requirements

An individual entering this course should have any of the following minimum requirements:

- (i) Kenya Certificate of Secondary Education (K.C.S.E.) with a minimum mean grade of D (Plain)

Or

- (ii) Nautical Science KNQF Level 4 or related qualification

And/or

- (iii) Equivalent qualifications as determined by TVET Authority

Credit Accumulation, Transfer and Exemptions

BMA guidelines on credit accumulation, transfer and exemptions shall apply.

Trainer qualification

A trainer for this course shall have a higher relevant qualification than KNQF level 5 and Regulator requirements must have relevant licenses and experience where applicable.

Assessment

The course will be assessed formatively and summatively:

- a) During formative assessment all performance criteria shall be assessed based on performance criteria weighting.
- b) Summative assessment shall focus on critical aspects of the Unit of competency.
- c) During summative assessment basic and common units shall be integrated or assessed concurrently with the core units.

- d) Formative and summative assessment weights shall constitute 60% and 40% of the overall score respectively.
- e) Theory and practical weight shall be 40:60 respectively for each unit of learning;
- f) For a candidate to be declared competent in a unit of competency, the candidate must meet the following conditions:
 - i) Obtained at least 40% in theory assessment in formative and summative assessments.
 - ii) Obtained at least 50% in practical assessment in formative and summative assessment where applicable.
 - iii) Obtained at least 50% in the weighted results between formative assessment and summative assessment where the former constitutes 60% and the latter 40% of the overall score.
- g) Assessment performance rating for each unit of competency shall be as follows:

MARKS	COMPETENCE RATING
80 -100	Mastery
65 - 79	Proficiency
50 - 64	Competent
49 and below	Not Yet Competent
Y	Assessment Malpractice/irregularities

RPL Assessment

Assessment for Recognition of Prior Learning (RPL) shall be conducted in line with National RPL requirements and guidelines as well as BMA RPL policy.

Certification

A candidate will be issued with a **partial certificate of competency** on demonstration of competence in a given unit of core unit of competency. To attain **full certification in Marine Engineering level 5** with **respective transcripts**, the candidate must demonstrate competence in all the units of learning as given in this qualification pack. The certificate will be issued by **Bandari Maritime Academy**

Licensing

The graduate to be issued with the Certificate of Proficiency as an Able Seafarer Deck as required under regulation II/5 of the STCW Convention and Merchant Shipping (Training and Certification) Regulations as amended, the graduate will be required to meet the following additional requirements to be issued by the regulator

- Complete at least 12 months approved seagoing service while qualified to serve as a rating forming part of a navigational watch,
- Complete the applicable STCW basic and advanced safety courses.
- Seafarer medical certification
- Meet other requirements as may be required from time to time under the STCW Convention



BASIC UNITS OF LEARNING

COMMUNICATION SKILLS

UNIT CODE: 0031 441 01A

Relationship to Occupational Standards

This unit addresses the unit of competency: Demonstrate communication skills

Duration of Unit: 30 hours

Unit Description

This unit describes the competencies required to use specialized communication skills to meet specific needs of internal and external clients, conduct interviews, facilitate discussion with groups and contribute to the development of communication strategies.

Summary of Learning Outcomes

1. Meet communication needs of clients and colleagues
2. Contribute to the development of communication strategies
3. Conduct interviews
4. Facilitate group discussions
5. Represent the organization

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery method	Duration (Hours)
1. Meet communication needs of clients and colleagues	• Communication process • Modes of communication • Medium of communication • Effective communication • Barriers to communication • Flow of communication • Sources of information • Organizational policies • Organization requirements for	• Interview • Role playing • Observation • Viewing of related videos	10

Learning Outcome	Content	Suggested delivery method	Duration (Hours)
	written and electronic communication methods • Report writing • Effective questioning techniques (clarifying and probing) • Workplace etiquette • Ethical work practices in handling communication • Active listening • Feedback • Interpretation • Flexibility in communication		
2. Contribute to the development of communication strategies	• Dynamics of groups • Styles of group leadership • Openness and flexibility in communication • Communication skills relevant to client groups	• Interview • Role playing • Observation • Viewing of related videos	5
3. Conduct interviews	• Types of interviews • Establishing rapport • Facilitating resolution of issues • Developing action plans	• Interview • Role playing • Observation • Viewing of related videos	5

Learning Outcome	Content	Suggested delivery method	Duration (Hours)
4. Facilitate group discussions	• Identification of communication needs • Dynamics of groups • Styles of group leadership • Presentation of information • Encouraging group members' participation • Evaluating group communication strategies	• Interview • Role playing • Observation • Viewing of related videos	5
5. Represent the organization	• Presentation techniques • Development of a presentation • Multi-media utilization in presentation • Communication skills relevant to client groups	• Interview • Role playing • Observation • Viewing of related videos	5

Suggested assessment method

- Observation
- Written

Recommended Resources

- Desktop computers/laptops
- Internet connection
- Projectors
- Telephone

INFORMATION TECHNOLOGY AND COMMUNICATION SKILLS

UNIT CODE: 0611 441 02A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Demonstrate digital literacy

Duration of Unit: 50 hours

Unit Description

This unit describes competencies required to use a computer and other digital devices for the purposes of communication, work performance and management at the workplace.

Summary of Learning Outcomes

1. Identify computer software and hardware
2. Apply security measures to data, hardware, software in automated environment
3. Apply computer software in solving tasks
4. Apply internet and email in communication at workplace
5. Apply desktop publishing in official assignments
6. Prepare presentation packages

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery method	Duration
1. Identify computer hardware and software	• Concepts of ICT • Functions of ICT • History of computers • Components of a computer • Classification of computers	• Instructor led facilitation of theory • Demonstration by trainer • Practical work by trainee • Viewing of related videos • Project • Group discussions	5
2. Apply security measures to data, hardware and software	• Data security and control • Security threats and control measures	• Instructor led facilitation of theory • Demonstration by trainer	10

Learning Outcome	Content	Suggested delivery method	Duration
	- Types of computer crimes - Detection and protection against computer crimes - Laws governing protection of ICT	- Practical work by trainee - Viewing of related videos - Project - Group discussions	
3. Apply computer software in solving tasks	- Operating system - Word processing - Spread sheets - Data base design and manipulation - Data manipulation, storage and retrieval	- Instructor led facilitation of theory - Demonstration by trainer - Practical work by trainee - Viewing of related videos - Project - Group discussions	7
4. Apply internet and email in communication at workplace	- Computer networks - Network configurations - Uses of internet - Electronic mail (e-mail) concept	- Instructor led facilitation of theory - Demonstration by trainer - Practical work by trainee - Viewing of related videos - Project - Group discussions	8
5. Apply desktop publishing in official assignments	- Concept of desktop publishing - Opening publication window - Identifying different tools and tool bars	- Instructor led facilitation of theory - Demonstration by trainer	10

Learning Outcome	Content	Suggested delivery method	Duration
	• Determining page layout • Opening, saving and closing files • Drawing various shapes using DTP • Using colour pellets to enhance a document • Inserting text frames • Importing and exporting text • Object linking and embedding • Designing of various publications • Printing of various publications	• Practical work by trainee • Viewing of related videos • Project • Group discussions	
6. Prepare presentation packages	• Types of presentation packages • Procedure of creating slides • Formatting slides • Presentation of slides • Procedure for editing objects	• Instructor led facilitation of theory • Demonstration by trainer • Practical work by trainee • Viewing of related videos • Project • Group discussions	10

Suggested Assessment Methods

- Oral questioning
- Observation
- Oral presentation
- Written report
- Project

Recommended Resources

- Desk top computers
- Laptop computers
- Other digital devices
- Printers
- Storage devices
- Internet access
- Computer software



ENTREPRENEURIAL SKILLS

UNIT CODE: 0413 441 03A

Relationship to occupational standards

This unit addresses the unit of competency: Demonstrate entrepreneurial skills

Duration of unit: 50 hours

Unit description

This unit describes the competencies critical to demonstration of entrepreneurial aptitudes. It involves, developing business innovation strategies, developing new markets, customer base, expanding employed capital and undertaking regional/county expansion while retaining motivated staff.

Summary of Learning Outcomes

1. Develop business innovation strategies
2. Develop new products/ markets
3. Expand customers and product lines
4. Motivate all staff/workers
5. Expand employed capital base
6. Undertake regional/county business expansion

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery methods	Duration
1. Develop business Innovation strategies	• Innovation in business • Business innovation strategies • Creativity for business development • New technologies in entrepreneurship • Linkages with other entrepreneurs • Setting strategic directions • New ideas and approaches • Entrepreneurial skills development • Market trends	• Instructor led facilitation of theory • Demonstration by trainer • Practice by trainee • Role play • Case study	10

Learning Outcome	Content	Suggested delivery methods	Duration
	- Monitoring and anticipating market trends - Products and processes in entrepreneurship - Business conventions and exhibitions - Business growth refocus		
2. Develop new products/ markets	- Feasibility study for new products - Identifying new sources of raw material and resources - New target markets/customers - Increasing products and services - Marketing improvement - Intrapreneurship and business growth	- Instructor led facilitation of theory - Demonstration by trainer - Practice by trainee - Role play - Case study	4
3. Expand customers and product lines	- Market demand - Regulatory environment - Creating product and services competitive advantages - Creating royal client base - Identifying and maintain new customers and markets - Advance product/ service promotions - Advance market expansion	- Instructor led facilitation of theory - Demonstration by trainer - Practice by trainee - Role play - Case study	6

Learning Outcome	Content	Suggested delivery methods	Duration
	• Small business records management • Book keeping and auditing for small businesses • Computer application software and programmes • ICT in customer and product diversification		
4. Motivate staff/workers	• Motivation of workers • `Communication at workplace for motivation purpose • Problem solving • Conflict resolution at place of work • Good staff/workers relation • Team building and team work • Staff development and enhancement • Culture of continuous improvement	• Instructor led facilitation of theory • Demonstration by trainer • Practice by trainee • Role play • Case study	8
5. Expand employed capital base	• Employed capital in business • Business share holdings • Types of shares • Shares diversification • Role of shareholders • Entrepreneurship • Increasing products and services	• Instructor led facilitation of theory • Demonstration by trainer • Practice by trainee • Role play • Case study	12
6. Undertake county/ regional	• Region/ county identification process	• Instructor led facilitation of theory	10

Learning Outcome	Content	Suggested delivery methods	Duration
business expansion	- Regional/ county laws and regulation - Business regional/county expansion - Regional/ County business expansion - Innovation in business - Business expansion and diversification - Resources for regional/county expansion - Small business Strategic Plan - Computer software in business development - ICT and business growth	- Demonstration by trainer - Practice by trainee - Role play - Case study	

Suggested assessment methods

- Observation
- Case studies
- Individual/group assignments
- Projects
- Written
- Oral

Recommended Resources

- Case studies for small businesses
- Business plan templates
- Laptop/ desktop computers
- Internet
- Telephone
- Writing materials

EMPLOYABILITY SKILLS

UNIT CODE: 0031 441 04A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Demonstrate employability skills

Duration of Unit: 50 hours

Unit Description

This unit covers competencies required to demonstrate employability skills. It involves conducting self-management, demonstrating interpersonal communication, critical safe work habits, leading a workplace team, planning and organizing work, maintaining professional growth and development, demonstrating workplace learning, problem solving skills and managing ethical performance.

Summary of Learning Outcomes

1. Conduct self-management
2. Demonstrate interpersonal communication
3. Demonstrate critical safe work habits
4. Lead small teams
5. Plan and organize work
6. Maintain professional growth and development
7. Demonstrate workplace learning
8. Demonstrate problem solving skills
9. Demonstrate workplace ethics

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery method	Duration
1. Conduct self-management	• Self-awareness • Formulating personal vision, mission and goals • Strategies for overcoming life challenges • Emotional intelligence • Assertiveness versus aggressiveness • Expressing personal thoughts, feelings and beliefs • Developing and maintaining high self-esteem	• Instructor lead facilitation of theory • Demonstrations • Simulation/Role play • Group Discussion • Presentations • Projects • Case studies	5

Learning Outcome	Content	Suggested delivery method	Duration
	- Developing and maintaining positive self-image - Articulating ideas and aspirations - Accountability and responsibility - Good work habits - Self-awareness - Self-development - Financial literacy - Healthy lifestyle practices		
2. Demonstrate interpersonal communication	- Meaning of interpersonal communication - Listening skills - Types of audience - Writing skills - Reading skills - Meaning of empathy - Understanding customers' needs - Establishing communication networks - Sharing information	- Instructor lead facilitation of theory - Demonstrations - Simulation/Role play - Group Discussion - Presentations - Projects - Case studies	7
3. Demonstrate critical safe work habits	- Stress and stress management - Punctuality and time consciousness - Leisure - Integrating personal objectives into organizational objectives - Resources utilization - Setting work priorities - HIV and AIDS - Drug and substance abuse - Handling emerging issues	- Instructor lead facilitation of theory - Demonstrations - Simulation/Role play - Group Discussion - Presentations - Projects - Case studies	5
4. Lead a small team	- Leadership qualities - Team building	Instructor lead facilitation of theory	5

Learning Outcome	Content	Suggested delivery method	Duration
	• Determination of team roles and objectives • Team performance indicators • Responsibilities in a team • Forms of communication • Complementing team activities • Gender and gender mainstreaming • Human rights • Maintaining relationships • Conflicts and conflict resolution	• Demonstrations • Simulation/Role play • Group Discussion • Presentations • Projects • Case studies	
5. Plan and organize work	• Functions of management ✓ Planning ✓ Organizing • Time management • Decision making process • Task allocation • Evaluating work activities • Resource utilization • Problem solving • Collecting and organising information	• Instructor lead facilitation of theory • Demonstrations • Simulation/Role play • Group Discussion • Presentations • Projects • Case studies	5
6. Maintain professional growth and development	• Opportunities for professional growth • Assessing training needs • Licenses and certifications for professional growth and development • Pursuing personal and organizational goals • Identifying work priorities • Recognizing career advancement	• Instructor lead facilitation of theory • Demonstrations • Simulation/Role play • Group Discussion • Presentations • Projects • Case studies	6
7. Demonstrate workplace learning	• Managing own learning • Contributing to the learning community at the workplace	• Instructor lead facilitation of theory • Demonstrations	5

Learning Outcome	Content	Suggested delivery method	Duration
	• Cultural aspects of work • Variety of learning context • Application of learning • Safe use of technology • Identifying opportunities • Generating new ideas • Workplace innovation • Performance improvement • Handling emerging issues • Future trends and concerns in learning	• Simulation/Role play • Group Discussion • Presentations • Projects • Case studies	
8. Demonstrate problem solving skills	• Problem identification • Problem solving • Application of problem-solving strategies • Resolving customer concerns	• Instructor lead facilitation of theory • Demonstrations • Simulation/Role play • Group Discussion • Presentations • Projects • Case studies	6
9. Demonstrate workplace ethics	• Meaning of ethics • Ethical perspectives • Principles of ethics • Values and beliefs • Ethical standards • Organization code of ethics • Common ethical dilemmas • Organization culture • Corruption, bribery and conflict of interest • Privacy and data protection • Diversity, harassment and mutual respect • Financial responsibility/accountability • Etiquette	• Instructor lead facilitation of theory • Demonstrations • Simulation/Role play • Group Discussion • Presentations • Projects • Case studies	6

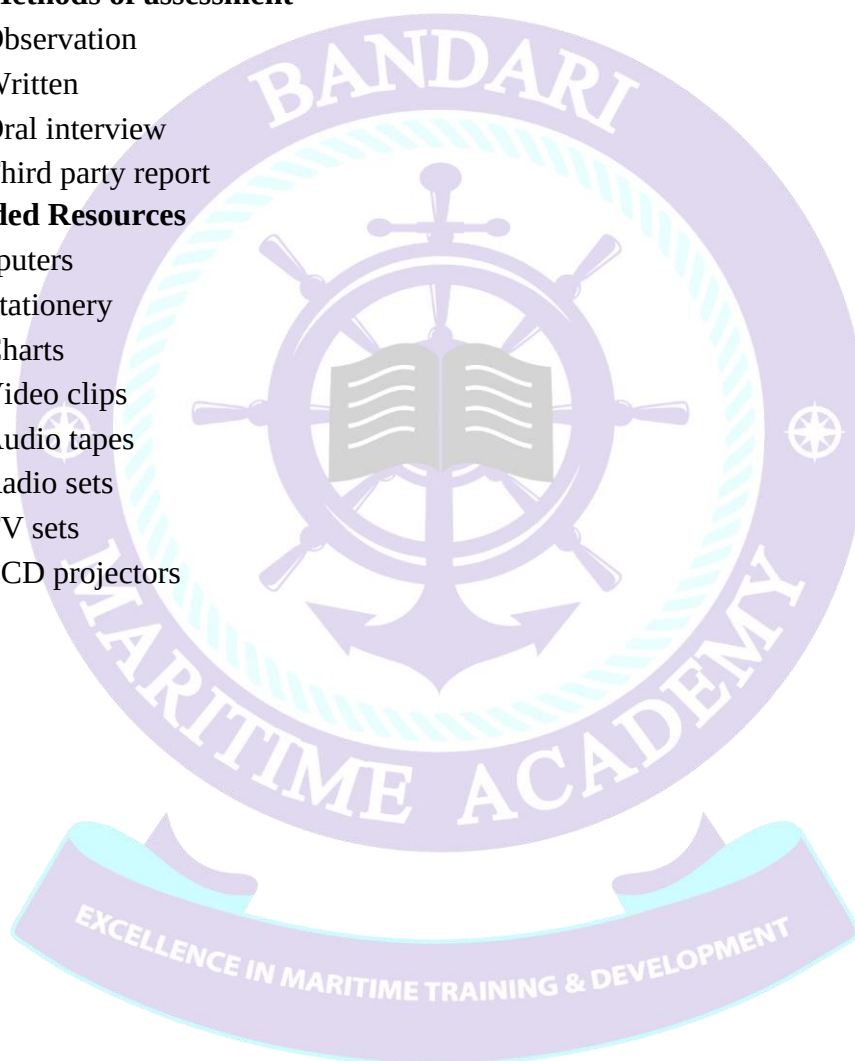
Learning Outcome	Content	Suggested delivery method	Duration
	• Personal and professional integrity • Commitment to jurisdictional laws • Emerging issues in ethics		

Suggested Methods of assessment

- Observation
- Written
- Oral interview
- Third party report

Recommended Resources

- Computers
- Stationery
- Charts
- Video clips
- Audio tapes
- Radio sets
- TV sets
- LCD projectors



OCCUPATIONAL SAFETY AND HEALTH PRACTICES

UNIT CODE:1022 441 05A

Relationship to Occupational Standards

This unit addresses the unit of competency: Demonstrate occupational safety and health practices

Duration of Unit: 30 hours

Unit Description

This unit describes the competencies required to comply with regulatory and organizational requirements for occupational safety and health.

Summary of Learning Outcomes

1. Identify workplace hazards and risk
2. Identify and implement appropriate control measures to hazards and risks
3. Implement OSH programs, procedures and policies/guidelines

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery method	Duration
1. Identify workplace hazards and risks	• Identification of hazards in the workplace and/or the indicators of their presence • Evaluation and/or work environment measurements of OSH hazards/risk existing in the workplace is conducted by • Authorized personnel or agency • Gathering of OHS issues and/or concerns raised	• Instructor led facilitation of theory • Demonstration by trainer • Practical work by trainee • Viewing of related videos	10
2. Identify and implement appropriate control measure to	• Prevention and control measures, including use of PPE (personal protective equipment) for specific hazards are	• Instructor led facilitation of theory • Demonstration by trainer	11

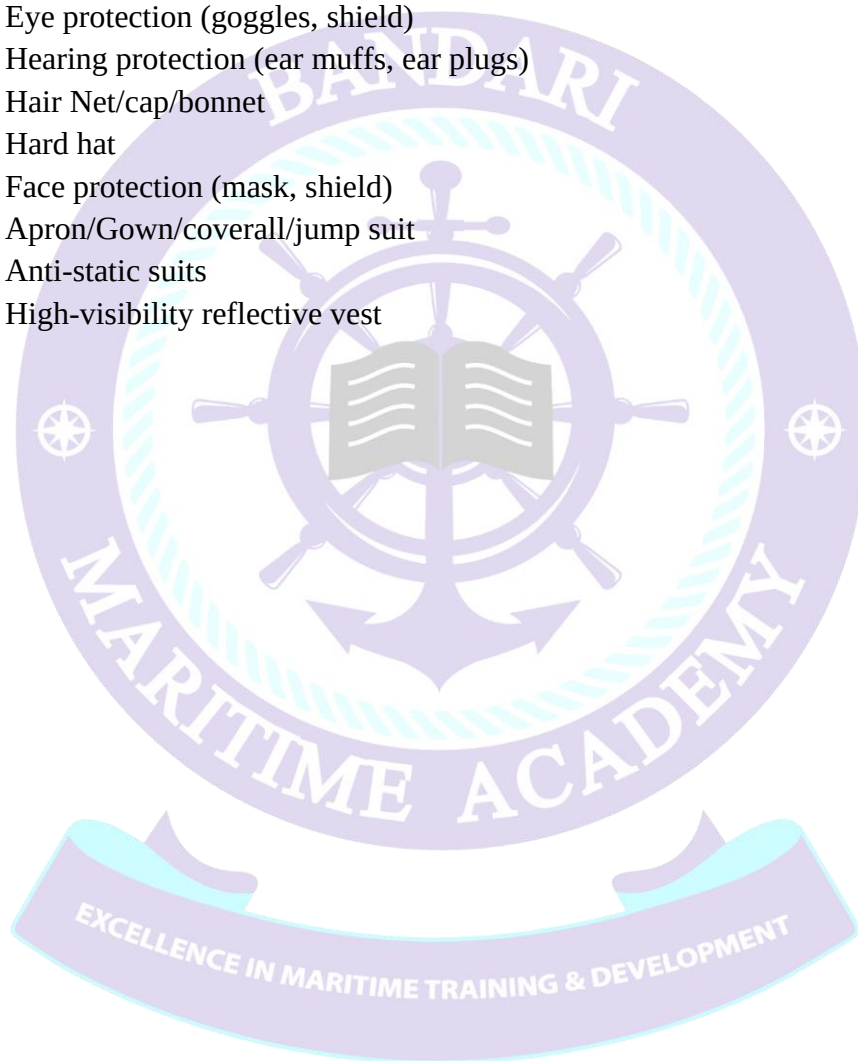
Learning Outcome	Content	Suggested delivery method	Duration
hazards and risks	identified and implemented • Appropriate risk controls based on result of OSH hazard evaluation is recommended • Contingency measures, including emergency procedures during workplace incidents and emergencies are recognized and established in accordance with organization procedures	• Practical work by trainee • Viewing of related videos	
3. Implement OSH programs, procedures and policies/guidelines	• Providing information to work team about company OHS program, procedures and policies/guidelines • Participating in implementation of OSH procedures and policies/guidelines • Training of team members and advice on OSH standards and procedures • Implementation of procedures for maintaining OSH-related records	• Instructor led facilitation of theory • Demonstration by trainer • Practical work by trainee • Viewing of related videos	9

Suggested Delivery Methods

- Oral questions
- Written tests
- Practical test
- Observation

Recommended Resources

- Standard operating and/or other workplace procedures manuals
- Specific job procedures manuals
- Machine/equipment manufacturer's specifications and instructions
- Personal Protective Equipment (PPE) e.g.
 - ✓ Mask
 - ✓ Face mask/shield
 - ✓ Safety boots
 - ✓ Safety harness
 - ✓ Arm/Hand guard, gloves
 - ✓ Eye protection (goggles, shield)
 - ✓ Hearing protection (ear muffs, ear plugs)
 - ✓ Hair Net/cap/bonnet
 - ✓ Hard hat
 - ✓ Face protection (mask, shield)
 - ✓ Apron/Gown/coverall/jump suit
 - ✓ Anti-static suits
 - ✓ High-visibility reflective vest



COMMON UNITS OF LEARNING





MARITIME MATHEMATICS

UNIT CODE: 1041 451 06A

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Maritime mathematics

Duration of Unit: 80 hours

Unit Description

This unit describes the competencies required by Deck Officer to apply a wide range of maritime mathematics in their work. It involves applying algebraic functions, application of trigonometry and hyperbolic functions, applying complex numbers, coordinate geometry, carrying out binomial expansion, calculus, Statistics, Vector theory, Matrix and Numerical methods in solving problems, probability, commercial calculations, performing estimations, measurements and calculation of quantities.

Summary of Learning Outcomes

1. Apply Algebra
2. Apply coordinate geometry
3. Apply trigonometry functions
4. Carry out mensuration
5. Apply statistics
6. Apply matrix
7. Apply vectors

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery methods	Duration
1. Apply Algebra	• Base and Index • Law of indices • Indicial equations • Laws of logarithm • Logarithmic equations • Conversion of bases • Use of calculator • Reduction of equations	• Group discussions • Demonstration by trainer • Facilitator led theory • Industrial visits	14

Learning Outcome	Content	Suggested delivery methods	Duration
	• Solution of equations reduced to quadratic form • Solutions of simultaneous linear equations in two unknowns • Solutions of problems involving AP and GP		
2. Apply Coordinate Geometry	• Polar equations • Cartesian equation • Graphs of polar equations • Normal and tangents • Definition of a point • Locus of a point in relation to a circle • Loci of points for given mechanism	• Group discussions • Demonstration by trainer • Facilitator led theory • Industrial visits	10
3. Apply trigonometric functions	• Half -angle formula • Factor formula • Trigonometric functions • Solutions of triangles ✓ Sine rule ✓ Cosine rule • Parametric equations • Relative and absolute measures • Measures calculation	• Group discussions • Demonstration by trainer • Facilitator led theory • Industrial visits	12

Learning Outcome	Content	Suggested delivery methods	Duration
	• Meaning of hyperbolic equations • Properties of hyperbolic functions • Evaluations of hyperbolic functions • Hyperbolic identities • Osborne's Rule • $Ashx + bshx = C$ equation		
4. Carry out mensuration	• Units of measurements and their symbols • Conversion of units of measurement • Calculation of length, width, height, perimeter, area, surface area, volume and angles of figures • Measuring tools and equipment • Performing measurements and estimations of quantities	• Group discussions • Demonstration by trainer • Facilitator led theory • Industrial visits	10
5. Apply Statistics	• Classification of data ✓ Grouped data ✓ Ungrouped data • Data collection • Tabulation of data ✓ Class intervals	• Group discussions • Demonstration by trainer • Facilitator led theory • Industrial visits	14

Learning Outcome	Content	Suggested delivery methods	Duration
	✓ Class boundaries ✓ Frequency tables • Diagrammatic and graphical presentation of data e.g. ✓ Histograms ✓ Frequency polygons ✓ Bar charts ✓ Pie charts ✓ Cumulative frequency curves □ Measures of central tendency mean, mode and median • Measures of dispersion ✓ Variance ✓ standard deviation • Definition of probability • Laws of probability • Expectation variance and S.D. • Types of distributions • Mean, variance and SD of probability distributions • Application of probability distributions		
6. Apply Matrix methods	• Matrix operation ✓ Addition ✓ Subtraction ✓ multiplication	• Group discussions	10

Learning Outcome	Content	Suggested delivery methods	Duration
	• Determinant of 2x2 matrix • Inverse of 2x2 matrix • Solutions of linear simultaneous equations in 2 unknowns • Application of matrices	• Demonstration by trainer • Facilitator led theory • Industrial visits	
7. Apply Vectors	• Definition of dot and cross product of vectors • Solution of problems involving dot and cross production of cross • Definition of operators • Definition of vector field • Solutions of problems involving vector fields • Definition of Gradient, Divergence and curl • Solutions involving Gradient, Divergence and curl • Application of vectors	• Group discussions • Demonstration by trainer • Facilitator led theory • Industrial visits	10

Suggested Methods of assessment

- Written tests
- Oral questioning
- Assignments

- Supervised exercises

Recommended Resources

- Scientific Calculators
- Rulers, pencils, erasers
- Charts with presentations of data
- Graph books
- Dice
- Computers with internet connection



MARINE ELECTRICAL AND ELECTRONICS PRINCIPLES

UNIT CODE: 1041 451 07A

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply principles of marine electrotechnology principles

Duration of Unit: 70 hours

Unit Description

This unit describes the competencies required to apply a wide range of marine electrotechnology principles. It involves use of the concept of basic Electrical quantities, use of the concepts of D.C and A.C circuits in electrical installation, use of basic electrical machine, use of power factor in electrical installation, use of earthing in Electrical installations, apply lightning protection measures, apply electromagnetic field theory and apply electrodynamics

Summary of Learning Outcomes

1. Use the concept of basic Electrical quantities
2. Use the concepts of D.C and A.C circuits in electrical installation
3. Use of basic electrical machine
4. Use of power factor in electrical installation
5. Use of earthing in Electrical installations
6. Apply lightning protection measures
7. Apply Electromagnetic field theory
8. Apply Electrodynamics

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery methods	Duration
1. Use the concept of basic Electrical quantities	• The meaning of SI unit • SI unit of various types of Electrical parameters • Ohm's law • Capacitors and inductors • Calculations involving various Electrical parameters e.g. Power, Current, Voltage, Resistance • Instruments used in measuring various types of Electrical parameters	• Written tests • Oral questioning • Assignments • Supervised exercises	10 hours

Learning Outcome	Content	Suggested delivery methods	Duration
2. Use the concepts of D.C and A.C circuits in electrical installation	• Meaning of terms • AC and DC, parallel and series circuits • AC and DC networks • R-L-C transients • AC to DC and DC to AC Conversion • Basic solar photovoltaic systems • Delta star conversion • Definition of power factor • Power triangle • Power factor correction	• Written tests • Oral questioning • Assignments • Supervised exercises	10 hours
3. Use Two Port networks	• Meaning passive networks • Types of Passive network • Characteristic impedance in T & pie networks • Design of T & pie networks • Transmission lines • ABCD Constants • Network in cascade	• Assignments • Oral questioning • Supervised exercises • Written tests	10 hours
4. Use of basic electrical machine	• Types of Electrical machines • DC machines, • AC Single and three phase motors, generators and Transformers • Application of AC and DC machines • Special machines and their applications	• Assignments • Oral questioning • Supervised exercises • Written tests • Practical tests	10 hours
5. Use of earthing in Electrical installations	• Definition of terms in earthing • Earthing points in Electrical installation • Types of earthing systems	• Assignments • Supervised exercises • Written tests • Practical test	10 hours

Learning Outcome	Content	Suggested delivery methods	Duration
	- Factors to consider in selecting an earthing method - Testing an earthing system		
6. Apply electrical system protection	- Electrical faults and abnormal conditions - Components of protection systems - Protective devices - Testing of protection systems - Coordination of protection systems	- Assignments - Oral questioning - Supervised exercises - Written tests	10 hours
7. Apply Electromagnetic field Theory	- Introduction to magnetism - Meaning of Electromagnetic Field Theory - Sources of Electromagnetic Fields - Detectors of Electromagnetic radiation - Application of Electromagnetic waves - Electromagnetics Laws - Faraday's Law - Lenz's law - Fleming's Laws - Properties and Effects of Electromagnetic waves - Wave Characteristics and Shielding - Skin Effect - Energy conservation theorem: - Poyntings' Theorem - Momentum Energy Flow	- Assignments - Oral questioning - Supervised exercises - Written tests	10 hours

Learning Outcome	Content	Suggested delivery methods	Duration
	Electromagnetic Energy flow		
8. Apply Electrodynamics	- Meaning of Electrostatics - Identification of Electrostatic terms and their meaning - Meaning of terms in magnetostatics - Electrodynamics laws	- Assignments - Oral questioning - Supervised exercises - Written tests	10 hours

Suggested Delivery Methods

- Group discussions
- Demonstration by trainer
- Exercises by trainee

Recommended Resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Scientific Calculators	25 pcs
2.	Relevant reference materials	25 copies
3.	Stationeries	25
4.	Electrical workshop	1
5.	Relevant practical materials	25
6.	Dice	25
7.	Computers with internet connection	25

EXCELLENCE IN MARITIME TRAINING & DEVELOPMENT

SHIP CONSTRUCTION PRINCIPLES

ISCED UNIT CODE: 1041 451 08A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Ship Construction principles

Duration of Unit: 70 hours

Unit Description

This unit describes the competencies required by a nautical operations officer to identify ship dimension and forms, interpret ship's drawings and plans, identify structural components of a ship, identify structural components of bow and stern, identify ship fittings, identify types of rudders and propeller, identify load lines and draught marks and identify ship stresses.

Summary of Learning Outcomes

1. Identify ship dimension and forms;
2. Identify structural components of a ship;
3. Identify structural components of bow and stern;
4. Identify ship fittings;
5. Identify types of rudders and propeller;
6. Identify load lines and draught marks;
7. Identify ship stresses
8. Check weather tight and watertight integrity of the ship

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Identify ship dimension and forms.	• Terms used to describe size and shape of vessels	• Observation • Project

Learning Outcome	Content	Suggested Assessment Methods
	• Terms used to describe key structural components of vessels • Terms used to describe size or cargo carrying capacity of vessels • Materials used in hull construction • Properties of high tensile steel • Types of steel used in hull construction • Use of forged, rolled and cast components in hull construction	• Written assessment • Oral assessment • Third party report • Interviews
2. Identify structural components of a ship.	• Purpose of bulkheads • Midship section of various types of ships • Bulkhead sketched showing construction and attachment to hull • Purpose, construction, and location of collision bulkheads • Test procedures for bulkheads • Maintaining strength in hull openings • Requirements for penetration of collision bulkhead • Structural design and construction for fire safety: – Bulkheads – Fire doors • Longitudinal, transverse, and	• Observation • Project • Written assessment • Oral assessment • Third party report

Learning Outcome	Content	Suggested Assessment Methods
	combined framing • Transition methods from one frame type to another • Maintaining strength of frames when connecting to deck beams and other strength members	
3. Identify structural components of bow and stern	• Panting and pounding forces • Strengthening forepeak sections to resist panting and pounding forces • Anchor and cable arrangements in forepeak tank • Strength members in afterpeak sections • Different rudder support arrangements	• Observation • Project • Written assessment • Oral assessment • Third party report
4. Identify ship fittings.	• Construction and arrangement of ship fittings • Strength members required for deck machinery and fittings • Watertightness and weathertight integrity as required under the SOLAS convention. • Reducing stress a. Hatchways b. Door openings c. Forecastle d. Bridge structure e. Watertight doors	• Observation • Project • Written assessment • Oral assessment • Third party report

Learning Outcome	Content	Suggested Assessment Methods
	f. Gastight doors • Use of suction and discharge valves and fittings in the shell above and below the waterline • Bilge keels structure	
5. Identify types of rudders and propeller	• Propeller Construction • Rudder Construction • Principles of screw propulsion	• Observation • Project • Written assessment • Oral assessment • Third party report
6. Identify load lines and draught marks	• Vessel freeboard • Vessel load line marks • Vessel condition of assignment of freeboard.	• Observation • Written assessment • Project • Oral assessment • Third party report
7. Identify ship stresses.	• Forces acting on a vessel hull • Typical weight and shear force curves • Bending moments • Protection of the hull from corrosion	• Observation • Written assessment • Project • Oral assessment • Third party report
8. Check weather tight and watertight integrity of the ship	• Vessel structural elements critical to weathertightness and watertight integrity • Ship inspections to verify weathertightness and watertight integrity • Ship's load line survey and certification • Ships safety construction survey	•

Learning Outcome	Content	Suggested Assessment Methods
	and certification • Routine checks to confirm weathertight and watertight integrity of a ship • Defects and damage identification, correction and reporting • Effect of severe sea and weather conditions on ship weather tight and watertight integrity	

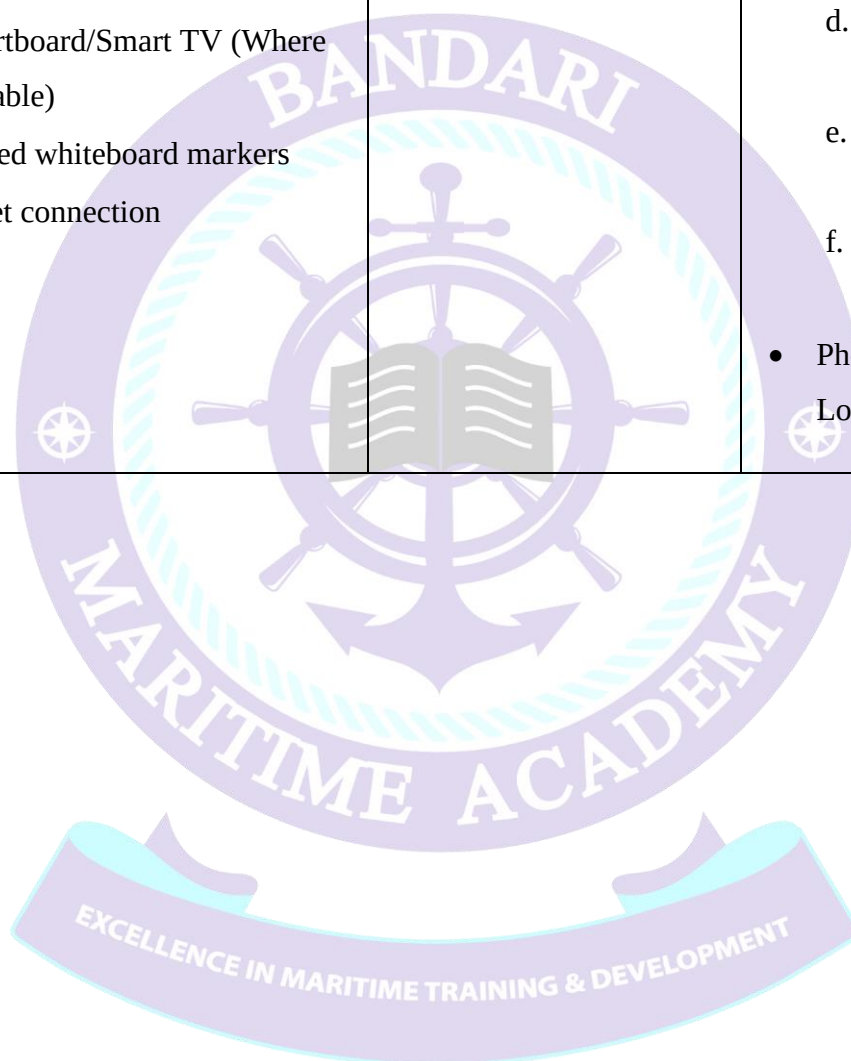
Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project
- Role play
- Case study

Recommended Resources for 24 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System	• Fully equipped seamanship workshop	• Load line convention • Ship plans • Stationery

b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Fully equipment marine engineering / Naval Architecture workshop	• Ship models and cut-outs • Photos of: a. Hatchways b. Door openings c. Forecastle d. Bridge structure e. Watertight doors f. Gastight doors • Photos of Loadline Marks
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SHIP STABILITY PRINCIPLES

UNIT CODE: 1041 451 10A

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply ship stability principles

Duration of Unit: 80 hours

Unit Description

This unit describes competencies required to apply ship stability principles. It involves monitoring hydrostatic data variation, demonstrating knowledge on buoyancy, applying fresh water allowance principles, applying statistical data, determining angle of loll, determining movement of a ship's centre of gravity and maintaining ship stresses within permissible limits

Summary of Learning Outcomes

1. Monitor hydrostatic variation
2. Demonstrate knowledge on ship buoyancy
3. Apply fresh water allowance principles
4. Apply statistical data
5. Determine angle of loll
6. Determine movement of a ship's centre of gravity
7. Apply damage control principles

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery methods	Duration
1. Monitor hydrostatic variation	• Principles of hydrostatic variation • Hydrostatic variation tests • Hydrostatic variation analysis • Tools, equipment and materials used in monitoring hydrostatic variation	• Demonstration by trainer • Practical work by trainee • Demonstration videos • Projects • Group discussions	20 hours
2. Demonstrate knowledge on ship buoyancy	• Principles of buoyancy • Factors affecting buoyancy • Apparent weight calculations • Relative density calculations	• Demonstration by trainer • Practical work by trainee • Demonstration videos • Projects • Group discussions	15 hours

Learning Outcome	Content	Suggested delivery methods	Duration
3. Apply fresh water allowance principles	• Draft measurement ○ Vessel draft in sea water ○ Vessel draft in fresh water • Measurement of fresh water allowance	• Demonstration by trainer • Practical work by trainee • Demonstration videos • Projects • Group discussions	14 hours
4. Determine angle of loll	• Definition of angle of loll • Vessel equilibrium condition checks • Metacentric heights ○ Negative ○ Positive • Corrective measures ○ Lowering center of gravity ○ Minimising free surfaces ○ Ballast tanks ○ Flooding and counter flooding ○ Load balancing	• Demonstration by trainer • Practical work by trainee • Demonstration videos • Projects • Group discussions	14 hours
5. Determine movement of a ship's centre of gravity	• Forces affecting ship's center of gravity ○ Identification ○ Analysis • Mass analysis ○ Effects of removing or discharging ○ Effects of loading or adding • Weight shifts ○ Horizontal ○ Vertical • Application of movement of ship's center of gravity calculation formulae	• Demonstration by trainer • Practical work by trainee • Demonstration videos • Projects • Group discussions	18 hours

Learning Outcome	Content	Suggested delivery methods	Duration
6. Maintain ship stresses within permissible limits	• Identification of stresses affecting ship stability • Analysis of stresses affecting ship stability • Permissible stress limits • Stress monitoring and maintenance	• Demonstration by trainer • Practical work by trainee • Demonstration videos • Projects • Group discussions	15 hours
7. Apply damage control principles	• Damage control principles • Damage control organisations ○ Shoring party ○ Leak stop party ○ Flooding and counter flooding party ○ Containment party ○ First Aid party • Damage control tools and equipment • Damage control plan • Damage control drills	• Demonstration by trainer • Practical work by trainee • Demonstration videos	14 hours

Suggested delivery methods

- Written tests
- Oral
- Practical/Projects

Recommended Resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Scientific Calculators	25 pcs
2.	Relevant reference materials	25 copies
3.	Stationeries	25
4.	Electrical workshop	1
5.	Relevant practical materials	25
6.	Dice	25
7.	Computers with internet connection	25

BASIC SEA SAFETY (STCW)

UNIT CODE: 1041 451 11A

Relationship to Occupational Standards

This unit addresses the unit of competency: **Apply Basic Sea Safety**

Unit duration: 50 hours

Unit description

This unit describes competencies required to apply basic sea safety. It involves applying Safe working practices, applying Personal Survival Techniques, carrying out Fire prevention & Firefighting, conducting Elementary First Aid, performing Personal Safety and Social Responsibility, and applying Maritime Security Awareness.

Summary of Learning Outcomes

1. Apply safe working practices
2. Apply Personal Survival Techniques
3. Carry out Fire prevention and Firefighting
4. Conduct Elementary First Aid
5. Perform Personal Safety and Social Responsibility
6. Apply Maritime Security Awareness

Learning Outcomes, Content, delivery method and duration

Learning Outcome	Content	Delivery method	Duration
1. Apply Safe working practices	• Introduction to Safe working practices • Working aloft & over the sides • Working in enclosed spaces • Permit to work systems • Line handling • Lifting techniques • Electrical safety • Mechanical safety • Chemical & biohazard safety • Personal Protective Equipment • Safety communications	• Instructor-led facilitation of theory • Videos • Group discussions • Trainer demonstrations • Practical	30 hrs

Learning Outcome	Content	Delivery method	Duration
2. Personal Survival Techniques	• Introduction to Basic Safety Training • Meaning of terms • Principles for survival at sea • SOLAS • Safety symbols • Emergency situations & Mustering • Ship evacuation • Personal Life Saving Appliances • Survival at sea • Emergency Radio Equipment • Helicopter assistance	• Instructor-led facilitation of theory • Videos • Group discussions • Practical	14 hrs
3. Fire Prevention and Firefighting	• Introduction to Fire Prevention and Firefighting • Fire safety and principles • Theory of Fire • Fire Prevention • Fire detection • Ship Firefighting Organization • Fixed Firefighting system • Firefighting equipment • Firefighting basics • Firefighting drills	• Instructor-led facilitation of theory • Videos • Group discussions • Trainer demonstrations • Practical	14 hrs
4. Elementary First Aid	• Introduction • Introduction to First Aid • Principles of First Aid • Body Structures and functions • Respiratory system	• Instructor-led facilitation of theory • Videos • Trainer demonstration • Group discussions	14 hrs

Learning Outcome	Content	Delivery method	Duration
	- Unconscious casualty - Cardio Pulmonary Resuscitation - Recovery position - Fracture - Spinal injuries - Shock treatment - Transfer of casualty	Practical	
5. Personal Safety & Social Responsibility	- Introduction - Emergency procedures - Protection of Marine environment - MARPOL - Safe working Practices - Lifting techniques - Permit to work systems - Effective communications onboard - Human relationships - Causes of Fatigue	- Instructor-led facilitation of theory - Group discussions - Group discussions - Practical	14 hrs
6. Maritime Security Awareness	- Introduction - Maritime security policy - Security Responsibilities - Threat Identification, Recognition and response - Ship security Actions - Emergency preparedness, Drills and Exercises	- Instructor-led facilitation of theory - Case studies - Group discussions - Videos	7 hrs

Suggested methods of assessment

- Oral questioning
- Written assessment
- Observation/practical

- Third party reports

Recommended resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Basic Safety Training handouts	25
2.	SOLAS Manual	5
3.	Swimming pool	1
4.	Diving pool	1
5.	Life jackets	25
6.	Immersion suits	25
7.	Work environment	1
8.	Life raft	1
9.	Fire simulator	1
10.	Smoke generator	2
11.	Complete fire suit	25
12.	Breathing apparatus	13
13.	Compressor	1
14.	Fire extinguishers	10
15.	Fire blankets	2
16.	Assorted stretchers	2
17.	Fixed firefighting systems	2

S/No.	Resource	Quantity (a class of 25 trainees)
18.	Messenger lines	2
19.	Overalls	25
20.	Medical kit	5
21.	Casualty model/dummy	2
22.	Assorted videos	10
23.	MARPOL manual	5
24.	VHF Radios	6
25.	Electro-mechanical safety manuals	5
26.	Safety communications manual	5
27.	Chemical/biohazard safety manuals	5
28.	Electrical workshop	1
29.	Mechanical workshop	1



METEOROLOGY

ISCED UNIT CODE: 0532 541 04A

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply meteorological practice

Duration: 60 hours

Unit description

This unit describes the competencies required by a nautical operations officer to apply meteorological practice. It involves applying basic concepts of metrology, reading shipborne meteorological instruments, recording weather information, identifying weather conditions, identifying sea and swell waves and applying basic principles of precipitation and cloud formation

Summary of Learning Outcomes

1. Apply basic concepts of metrology
2. Read shipborne meteorological instruments
3. Record weather information
4. Identify weather conditions
5. Identify sea and swell waves
6. Apply basic principles of precipitation and cloud formation

Learning Outcomes, content, and suggested assessment methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply basic concepts of metrology	• Introduction to metrology • Definition of terms • Clouds • visibility at sea • Pressure systems, • Wind systems, • Atmosphere • Cyclones and Anticyclones • Fronts	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
2. Read shipboard meteorological instruments	• Introduction to shipboard meteorological instruments • Shipboard meteorological instruments • Safe working practices in calibration and use of shipboard meteorological instruments • Safe working practices maintenance of shipboard meteorological instruments • weather information obtained from shipboard meteorological instruments	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Record weather information	• Introduction to recording weather information • weather information to be recorded • Meteorological observations, measurements and recording on board ships • Meteorological reports	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
4. Identify weather conditions	• Introduction to weather conditions • Factors considered to identify weather conditions – Temperature – Humidity – Precipitation – Wind – Cloud cover – Visibility – Air pressure – Weather events • Weather instruments – Dry and wet thermometer	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	– Aneroid Barometer – Barograph – Rain gauge – Anemometer ● Types of weather condition – Sunny/ clear – Partly cloudy – Overcast/cloudy – Rainy – Thunderstorms – Snowy – Sleet – Hail – Foggy – Windy – Humid – Hurricane – Typhoon – Cyclone – Tornado – Drizzle	
5. Identify sea and swell waves	● Introduction to sea swell and waves ● Surface water circulation ● Ocean current systems ● Nautical publications on tides and currents ● Calculation of tidal conditions. ● Forecasting wave height and swell ● Sea waves, swell, storm surge, tsunami and bore tides. ● wind rose.	● Observation ● Written assessment ● Oral assessment ● Third party reports ● Portfolio of evidence ● Project ● Practical

Learning Outcome	Content	Suggested Assessment Methods
6. Apply principles of precipitation and cloud formation	• Introduction to cloud formation and precipitation • Clouds and precipitation terminology • Main components and cloud formation processes • Clouds classification • Observation and recording of weather elements • Interpretation and recording of weather elements in navigation practice	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – Windows/Linux/Macintosh Operating System – Microsoft Office Software – Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector	• Dry and wet thermometer • Aneroid Barometer • Barograph • Rain gauge • Anemometer	• Weather charts • Company ISM manual • Tidal Charts • Beaufort Scale • Weather photos • Cloud Photos

General Resources	Tools and Equipment	Materials and Supplies
• 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection		



MARINE ENVIRONMENT PROTECTION

UNIT CODE: 1041 451 09A

Relationship to Occupational Standards

This unit addresses the unit of competency: Protect marine environment

Duration of Unit: 50 Hours

Unit Description

This unit specifies competencies required by a Rating to apply marine environment protection. It involves identifying MARPOL 73/78 requirements for protection of marine environment, applying measures to protect marine environment, apply operational principles of anti-pollution equipment, applying procedures for disposal of garbage, applying ballast water operations procedures, maintaining shipboard records, related to protection of marine environment, national legislation relevant to environmental protection.

Summary of Learning Outcomes

1. Identify MARPOL 73/78 requirements for protection of marine environment.
2. Apply precautions to protect marine environment
3. Apply operational principles of anti- pollution equipment
4. Apply procedures for disposal of garbage
5. Apply ballast water operation procedures
6. National legislation relevant to environmental protection.

Elements Learning Outcomes, Content and Suggested assessment methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Identify MARPOL 73/78 requirements for protection of marine environment.	• Meaning of MARPOL 73/78 • Terms used in MARPOL 73/78 • Requirements of MARPOL 73/78 in protection of marine environment: - Annex 1 prevention of pollution by oil - Annex 2 prevention of pollution by noxious liquid carried in bulk - Annex 3 prevention of pollution by harmful	• Observation • Written assessment • Project • Practical assessment

Learning Outcomes	Content	Suggested Assessment Methods
	substances carried in packaged form - Annex 4 prevention of pollution by sewage - Annex 5 prevention of pollution by garbage - Annex 6 prevention of pollution by air • Maintain shipboard records in accordance with legislation and company policy.	
2. Apply precautions to protect marine environment	• Shipboard measures to protect marine environment • Shipboard drills to protect marine environment as per company policy. • Identify potential non-compliance and reporting in accordance with organisational procedures. • Maintain shipboard records in accordance with legislation and company policy.	• Observation • Written assessment • Project • Practical assessment
3. Apply operational principles of anti-pollution equipment	• <i>Define anti-pollution equipment</i> • <i>Identify antipollution equipment</i> • <i>Operation of anti-pollution equipment as per company policy</i> • <i>Monitoring anti-pollution equipment in accordance with regulatory and organizational requirements.</i> • Maintain shipboard records in accordance with legislation and company policy.	• Observation • Written assessment • Project • Practical assessment

Learning Outcomes	Content	Suggested Assessment Methods
4. Apply procedures for disposal of garbage	- Types of garbage as per MARPOL 73/78 and company policy - Plastics - Cargo residue - Food waste - Cleaning agents and - Chemicals - Identify requirements for disposal of garbage as per MARPOL 73/78 and company policy. - Maintain shipboard records in accordance with legislation and company policy.	- Observation - Written assessment - Project - Practical assessment
5. Apply ballast water operation procedures	- Define ballast water - Apply Ballast Water Management Plan as per MARPOL 73/78 and national regulation - Operate ballast water management systems as per company policy. - Apply safety procedures for operation of ballast water management and treatment systems based on the company policy and manufacturer instructions. - Identify areas of compliance areas and inspection under port State control regime - Maintain shipboard records in accordance with legislation and company policy.	- Observation - Written assessment - Project - Practical assessment
6. National legislation relevant to protection of marine environment	Identify national legislation relevant to protection of marine environment	- Observation - Written assessment - Project - Practical assessment

Learning Outcomes	Content	Suggested Assessment Methods
	- Identify purpose of national legislation relevant to protection or marine environment - Implement national legislation on MARPOL 73/78 as per company policy. - Monitor shipboard operations for compliance with national legislation and company policy. - Report incidents as per national legislative requirements and company policy. - Maintain shipboard records in accordance with national legislation and company policy.	

Suggested Methods of Instruction

1. Group discussions
2. Self-directed learning
3. Demonstrations
4. Role play
5. Practical
6. Project
7. Oral presentations
8. Video presentations

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
24 computers with the following software: - Windows/Linux/Macintosh Operating System - Microsoft Office Software - Antivirus Software 2 Printers	- SOPEP Equipment - ODME - Oil Discharge Monitoring Equipment (Recommended if available)	Eight (8) copies of International Convention for the Prevention of Pollution from Ships (MARPOL) 73/78

General Resources	Tools and Equipment	Materials and Supplies
• Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Oil Water Separator (Recommended if available)	• Eight (8) copies of Ballast Water Management Convention (BWM), 2004) • Sample SOPEP SoP • Sample Company ISM manual • National marine spills contingency plan • Oil Dispersant Use Policy • Oil Spill Notification form • Merchant shipping notices • Industry circulars and recommendation



MARITIME COMMUNICATION SKILLS

ISCED UNIT CODE: 1041 451 13A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply maritime communication skills

Duration: 40 Hours

Unit Description

This unit covers the competencies required to demonstrate maritime communication skills. It involves, applying marine communication phrases, using English language in written and oral communication, using internal communication system onboard, executing commands in emergency situations, recognizing existence of cultural and religious diversities in the maritime sector and preparing shipboard report.

Summary of learning outcomes

1. Apply marine communication phrases
2. Use English in written and oral communication
3. Use internal communication systems onboard
4. Execute commands in emergency
5. Recognize the existence of cultural and religious diversities in the maritime sector
6. Prepare shipboard reports.

Learning outcomes, content, and suggested assessment methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply marine communication phrases	● Marine communication phrases a) Communication procedures: – Distress – Search and rescue – Radio medical advice – Safety communication – Urgency communication – Environmental protection – Meteorological and hydrological condition – Navigational warnings – Pilotage – Helicopter operations – Icebreaker operations – Vessel traffic service (VTS) standard phrases	● Observation ● Written assessment ● Oral assessment ● Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	b) Onboard communication phrase: – Standard helm orders – Standard engine orders – Pilot on the bridge – Handing over watch – Safety onboard – Cargo and cargo handling – Preparing for sea – Passenger care ● Marine communication challenges – Language barrier – Cultural differences – Noise – Communication equipment failure – Attitude – Lack clarity – Information overload – Expectations prejudices – Email overload ● Marine communication channels – Written communication – Verbal communication – Nonverbal communication	
2. Use English in written and oral communication	● Terms and concept used in communication ● Communication Process ● Classification of communication ● Forms of Communication ● Channels of Communication ● Writing Skills ● Summary ● Report Writing Skills ● Conducting Meetings and Minute Writing ● Interviews	● Observation ● Portfolio of Evidence ● Project ● Written assessment ● Practical assessment ● Oral assessment
3. Use internal communication	● Types of internal communication systems onboard	● Observation ● Portfolio of Evidence

Learning Outcome	Content	Suggested Assessment Methods
systems onboard	– GMDSS – Intercom – Public address – Telephones	• Project • Written assessment • Practical assessment • Oral assessment
4. Execute commands in emergency situations	• Types of emergencies onboard – Sinking – Man overboard – Fire – Grounding – Foundering – Flooding – Piracy – enclosed space rescue – Evacuation – Collision – Power black out • Emergency signals – Continuous ringing of ships bell – One long blast – General alarm- seven short and one long blast – Flare guns – Fire in a drum – Signaling flags – Pyrotechnics – May day call – Three long blasts	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment
5. Recognize the existence of cultural and religious diversities in the maritime sector	• Elements of interpersonal relationship onboard – Good company policies – Good shipboard management – Good technical competencies – Interpersonal relationship of all crew and officers • Effective communication skills	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Listening skills – People skills – Emotional intelligence – Appropriate skill selection – Ethical communication • Communication challenges – Language barrier – Cultural differences – Noise – Communication equipment failure – Attitude – Lack clarity – Information overload – Expectations prejudices – Email overload	
6. Prepare shipboard reports	• Meaning of report • Types of reports – Oral reports – Written – Management – Operations procedures – Production stages – Maintenance, break down and accident reports – Internal memos • Role of report in a company • Preparation for report writing – Audience analysis – Reading skills – Data collection – Data analysis	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Suggested methods instruction

- Blended/Hybrid learning
- Case study

- Coaching
- Cooperative learning (Think, pair, share, and Jigsaw)
- Demonstration by trainer
- Flipped classroom
- Group Discussions
- Industrials visits/ Industrial training practical's
- Inquiry-based Learning
- Instructor-led
- Journaling
- Practical Exercise by the trainee(s)
- Reflective discussion
- Roleplaying
- Self-study

Recommended resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – Windows/Linux/Macintosh Operating System – Microsoft Office Software – Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• 1 GMDSS Simulator station • Radar reflector • Portable VHF Radio communication • Shipboard Intercom • PA system • Talkback • Company ISM manual	• 2 Sets International code of signals • 1 Set Flags and shapes • 4 sets Pyrotechnics • 4 Copies of the IMO SMCP Resolution A.918(22) • 4 Copies of the IMO Resolution A.954 23) - Proper Use of VHF Channels at Sea • 4 Copies of SOLAS Chapter V Regulations: Safety of Navigation

CORE UNITS OF LEARNING



SEAMANSHIP PRACTICES

UNIT CODE: 1041 451 14A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform Seamanship Practices

Duration: 80 Hours

Unit Description

This unit specifies competencies required by Able Seafarer Deck to contribute to perform seamanship practices. It involves performing rope work and rigging, performing ship husbandry, operating ship steering systems, performing bilge, ballast and cargo pumping operations, implementing safe use and care for hand and power tools, implementing manufacturers safety guidelines and shipboard instructions, implementing procedures for safe disposal of waste materials, executing routine maintenance and repair procedures and executing maintenance and stowage of ropes, cables and chains

Summary of learning outcomes

1. Perform rope work and rigging
2. Perform ship husbandry
3. Operate ship steering systems
4. Perform Bilge, ballast and cargo pumping operations
5. Implement safe use and care for hand and power tools
6. Implement manufacturers safety guidelines and shipboard instructions
7. Implement procedures for safe disposal of waste materials
8. Execute routine maintenance and repair procedures
9. Execute maintenance and proper stowage of ropes, cables and chains

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Perform rope work and rigging	• Introduction to rope work and rigging • Meaning of terms • Health & safety at work precautions • Inspection and maintenance of ropes and gear • Risk assessment • PPE • Classification of ropes by material • Natural vs Synthetic fibres	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	• Classification of ropes by construction – Twisted – Braided – Kernmantle • Characteristics and application of ropes • Maintenance and care • Common knots and Hitches – Reef knot – Bowline – Double bow line – Fisherman's Bend – Carrick Bend – Figure-eight – Sheet Bend – Clove hitch – Rolling Hitch – Half hitch – Round turn and two half hitches • Splicing techniques – Eye splice – Short splice – Long splice – Back splice • Concepts of Rigging – Mechanical advantage – Center of gravity – Load distribution • Rigging equipment – Blocks and tackle – Shackles and hooks – Wire ropes – Winches and capstans • Load handling and safety – Static vs Dynamic – Safe working load – Breaking Strength	

Learning Outcome	Content	Suggested Assessment Methods
2. Perform Ship Husbandry	• Introduction to Ship Husbandry • Meaning of Terms • Health & safety at work precautions – PPE – Fire, safety and Deck rounds • Cleaning and Maintenance Techniques – Cleaning procedure – Maintenance Techniques • Surface preparation and painting – Surface preparation – Painting techniques • Minor repairs and Fabrication – Repairing damaged components – Repairing wooden and metal surfaces • Fabrication skills – Basic carpentry and metal work – Using hand and power tools	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment
3. Operate ships steering Systems	• Introduction to Ship steering systems • Health & safety at work precautions – PPE – Emergency stop procedures • Types of Steering Systems – Manual steering – Hydraulic steering – Electric and electro-hydraulic steering – Integrated bridge systems • Components of steering system – Helm – Rudder and Rudder Stock – Steering gear – Control systems and interface	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Operation of steering systems – Helm controls – Steering commands and response – Steering gear indicators • Routine inspection and maintenance of steering	
4. Perform Bilge, ballast and cargo pumping operations	• Health and safety at work precautions • Introduction to Bilge ballast and cargo pump operations • Types of pumping systems – Bilge Pumping System – Ballast pumping system – Cargo pumping system • Components of Bilge Pumping System – Bilge pumps – Bilge wells and strainer boxes – Pipework and valves • Operation of Bilge pumping systems – Pumping procedures – Monitoring • Components of ballast pumping – Ballast pumps and pipework – Ballast tanks and sensors – Valves • Operation – Ballast water treatment and regulations – Ballast water exchange procedure – Managing ballast for stability reasons. • Components of cargo pumping system	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Cargo pumps – Manifolds and piping system – Gauging and monitoring equipment • Operation – Cargo loading and discharging – Procedures for different type of cargo – Safety measures during operation	
5. Implement safe use and care for hand and power tools	• Introduction to safe use and care for hand and power tools • Health and safety at work precautions – PPE – Safe work environment setup – Emergency response measure • Principles of Tool safety – Types of hand and power tools – Common risks and hazards • Selecting hand tools – Right tool for right job • Using hand tools – Proper technique – Ergonomics and safe handling practices • Selecting power tools – Criteria for selecting right tool • Using Power tools – Proper technique – Safety precautions when operating power tools – Managing power cords and battery tools	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Maintenance Procedures – Routine inspection of hand tools – Lubrication, sharpening and cleaning – Identifying and repairing damaged tools • Storage practices – Organizing and storing tools – Using tool storage solutions	
6. Implement manufacturers safety guidelines and shipboard instructions	• Introduction to manufactures safety guidelines and shipboard instruction • Regulatory framework – International maritime safety standards (IMO, SOLAS, STCW) – Overview manufacturers safety guidelines and shipboard instructions • Components of safety Guidelines – Interpreting Material Safety Data sheet (MSDS) – Understanding equipment manuals and guidelines – Identify safety symbols and warnings • Application and implementation of safety guidelines • Components of Shipboard instructions – Understanding SOPs – Reading shipboard manuals – Roles and responsibilities in following instructions • Application of Shipboard instructions	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	- Integrating guidelines and instructions into daily operations - Routine inspections - Safe handling and storage of materials - Effective safety guidelines communication - Safety briefings and drills - Safety practices and emergency procedures - PPE correct use - Safe work environment setup - Emergency procedures for various incidents - Report and documenting incidents	
7. Implement procedures for safe disposal of waste materials	- Introduction to procedures for safe disposal of waste materials - Health and safety and environment procedures - PPE - Safe work environment set up - Waste related emergency response procedure - Minimize waste generation - Pollution prevention, response and containment - Types of waste - Solid waste (plastics, paper, food waste) - Liquid waste (bilge water, ballast water, sewage) - Hazardous waste (oil, chemicals, medical waste) - Regulatory framework - Marpol, - SOLAS and - STCW	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Compliance requirements – Record keeping and documentation, – Inspection and enforcement • Solid waste management Procedure – Segregation and storage – Recycling and reuse – Disposal methods • Liquid waste management Procedure – Treatment and disposal of bilge and ballast water – Sewage treatment and discharge procedure – Grey water discharge and handling • Hazardous waste Management – Identification and classification – Handling and disposal	
8. Execute routine maintenance and repair procedures	• Introduction to routine maintenance and repair procedures • Principles of maintenance – Preventive vs corrective maintenance – Maintenance schedules and log – Maintenance management systems • Tools and equipment – Hand tools – Power tools • Routine maintenance Procedures – Hull maintenance – Deck maintenance • Hull maintenance – Clean and inspect hull	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Identify, report and repair minor damage – Apply anti fouling coat • Deck Maintenance – Clean and maintain deck equipment – Inspect and repair safety equipment – Maintain deck fittings and structures	
9. Execute maintenance and proper stowage of ropes, cables and chains	• Introduction to maintenance and proper stowage of ropes cables and chains • Maintenance of ropes cables and chains – Inspection – Cleaning – Repair and replacement – lubrication • Inspection – Regular checks – Documentation • Cleaning – Ropes with fresh water – Cables remove rust with wire brush. Light coat of lubricant • Repair and replacement – Splicing on ropes for minor damage – End to end rotation for chains and cables – Replacement for chains, ropes and cables worn or damaged • Lubrication – Steel cables and – chains	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Factors to consider for proper Stowage of ropes, cables and chains – Storage areas – Cooling and flaking – Securing – Labelling and segregation • Storage areas – Dry and ventilated for ropes – Away from direct sunlight • Coiling and flaking – Coil ropes neatly to avoid tangling and kinks – Flake out cables and chains properly to avoid twists and knots • Securing – Tie downs – secure ropes with ties or lashings – Chain lockers-chain lockers or bins to store chain • Labelling and segregation – Identification –label ropes cables and chain with length type and use – Segregation- store ropes cables and chains separate to avoid damage.	

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project
- Role play
- Case study

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – Windows/Linux/Macintosh Operating System – Microsoft Office Software – Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Fully equipped seamanship workshop • Steering console/navigation simulator	• International handbook of Marine Navigation and Safety at Sea Transport by Adam Weintrit • Seamanship Techniques by D.J House • Bridge Procedures Guide by International Chamber of Commerce • COSWP Manual • Sample Company ISM Manual • Manila Ropes [various sizes] • 3 sets of synthetic Ropes of various sizes for each trainee • Steel wire Ropes of various sizes for each trainee • Seizing twine and seizing wire • 4 Heaving Lines • 8 Safety Belt • 8 Safety Harness • 8 Marline Spikes • 8 Wooden Spikes

WATCHKEEPING PRACTICES

UNIT CODE: 1041 451 15A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Watchkeeping Practices

Duration: 80 Hrs

Unit description

This unit specifies competencies required by Able Seafarer Deck to perform safe watchkeeping practice. It involves contributing to watchkeeping duties at sea, performing helmsmanship duties, performing lookout duties, performing watchkeeping duties at port in normal conditions, performing watchkeeping duties during cargo operations and performing watchkeeping duties at anchorage.

Summary of learning outcomes

1. Contribute to watchkeeping duties at sea
2. Perform Helmsmanship Duties
3. Perform Lookout Duties
4. Perform watchkeeping duties at port in normal conditions
5. Perform watchkeeping duties during cargo operations
6. Perform watchkeeping duties at anchorage

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Contribute to watchkeeping duties at sea	• Importance of watchkeeping at sea • Bridge Layout • Bridge Equipment Familiarization – Autopilot – Dual Radar/ARPA – Gyro – Position fixing systems – Dual ECDIS setup (Master + Backup) – Conning Display – Power distribution system	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Steering gear – GMDSS • Procedures of watch hand over and take over • Situational awareness • Winds, tides and current • Fire and safety rounds • Security and anti-piracy watch • Flags, Shapes and Signals	
2. Perform Helmsmanship Duties	• Duties of Helmsman • The Compass • The Steering Control System • Steering Modes – Follow Up, – Non-Follow Up • Maritime English commands used in Steering Midships – Port / starboard five – Port / starboard ten – Port / starboard fifteen – Port / starboard twenty – Port / starboard twenty-five – Hard-a-port / starboard – Nothing to port / starboard – Meet her – Steady – Easy to five / ten / fifteen / twenty – Steady as she goes • Basics of Vessel Behavior in Different Sea conditions/types of Ships • Steering in different conditions of visibility – Day and Night • Steering in Narrow Channels	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	(countering bank effect) - Steering in Shallow Waters (Sluggish steering). - Most effective steering under pilot-age - Action in case of emergency - Equipment failure – - Situational awareness - Man Overboard Maneuvers - Possible causes of error due to Human element	
3. Perform Lookout Duties	- The Bridge Team - Duties of a Lookout - Compass Bearings and reporting of targets - COLREGS - Rule 5 - Risk of Collision - Action to be taken to Avoid Collision - Lights and Shapes - Sounds and Light Signals - Buoyage System	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
4. Perform watchkeeping duties at port in normal conditions	- Mooring Layout - Tending to Mooring Lines - Watch Levels/ Security at Port - Gangway Watch - Security Rounds - Provisions - Rigging of Pilot Ladders and Gangway (Safe Access) - Ballasting/ deballasting Operations - Pollution Prevention at Port	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
5. Perform watchkeeping	- Drafts - Cargo Handling Equipment	- Observation - Portfolio of

Learning Outcome	Content	Suggested Assessment Methods
duties during cargo operations	• Loading/ offloading of cargo • Cargo Stowage and Securing • Duties during Cargo Watch	Evidence • Project • Written assessment • Practical assessment • Oral assessment
6. Perform watchkeeping duties at anchorage	• Anchor/ chain position and direction • Duties during Anchor Watch • Measures to protect the environment from pollution by the ship and comply with applicable pollution regulations. • Traffic conditions;	• Written assessment • Oral assessment • Practical assessment

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – Windows/Linux/Macintosh Operating System – Microsoft Office Software – Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard	• Fully equipped seamanship workshop • Steering console/ navigation simulator	• International handbook of Marine Navigation and Safety at Sea Transport by Adam Weinrit • Seamanship Techniques by D.J House • Bridge Procedures Guide by International

• 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection		Chamber of Commerce
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DECK MACHINERY, CARGO HANDLING GEAR EQUIPMENT OPERATION AND MAINTENANCE

UNIT CODE: 1041 451 16A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform safe operation and maintenance of deck machinery, cargo handling gear and equipment

Duration: 70 Hours

Unit Description

This unit specifies competencies required by an Able Seafarer to perform operation and maintenance of deck machinery, cargo handling gear and equipment. It involves implementing preparation procedures for operations of deck machinery, cargo handling gear and equipment, undertaking deck machinery operation, undertaking cargo handling gear and equipment operation, rigging and unrigging Bosun's chairs, pilot ladders, hoists, rat guards and gangways, operating anchoring equipment under various conditions, using signals for operations of equipment, implementing operations completion procedures, operating access arrangements and pipeline systems, maintaining mooring equipment and maintaining cargo handling gear and equipment.

Summary of learning outcomes

1. Implement preparation procedures for operations of the deck machinery, cargo handling gear and equipment
2. Undertake deck machinery operation
3. Undertake cargo handling gear and equipment operations
4. Rig and unrig Bosun's chairs staging, pilot ladders, hoists, rat guards and gangways
5. Operate anchoring equipment under various conditions
6. Use signals for the operations equipment
7. Implement operations completion procedure
8. Operate access arrangements and pipeline systems
9. Maintain mooring equipment
10. Maintain cargo handling gear and equipment
11. Maintain deck equipment

Learning Outcome	Content	Suggested Assessment Methods
1. Implement preparation procedures for operations of the deck machinery, cargo handling gear and equipment	• Health and Safety at Work Precautions: – Safety Procedures – Reporting – Safety Devices Provided on Board – Specific Risks: - i.) Mechanical Risks - ii.) Non-Mechanical Risks; • Stability of Work Equipment: – Use of Clamps – Potential Movement of a Ship Under All Conditions • Dangerous Parts of Work Equipment & Danger Zone: – Work Equipment Guards or Protection Devices – Maintenance of Guards and Protection Devices – Equipment Danger Zones • Lighting • Markings: - i.) Safe working load (SWL) - ii.) Expiry Dates - iii.) Risk Assessment Register • Start Up Procedure • Emergency Stop Procedure • Reporting	• Observation • Written assessment • Oral assessment • Practical assessment
2. Undertake deck machinery operation	• Introduction • Deck Machinery: – Portable Power-Operated Tools & Equipment – Workshop and Bench Machines – Abrasive Wheels – Hydraulic, Pneumatic/High-Pressure Jetting Equipment – Hydraulic Jacks – Mobile Work Equipment	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Self-Propelled Work Equipment – Remote Controlled Self-Propelled Work Equipment – Laundry Equipment – Ropes and Wires • Deck Machinery Inspection: – Initial Inspection – Visual Inspection – Monthly Inspection – Annual Inspection – 5 Yearly Inspection – Reporting • Function and Usage of: – Hatches, – Weather Tight Doors – Water Tight Doors and Ports • Fibre and Wire Ropes, Cables and Chains: – Their Construction, – Use of Fibre and Wire Ropes, Cables and Chains – Markings, – Maintenance, – Proper Stowage. • Basic Signals for The Operation of Equipment: – Basic Signals for Operating Winches, – Basic Signals for Operating Windlasses, – Basic Signals for Operating Hoists and Cranes. • Function and Uses: – Function and Uses of Winches, – Function and Uses of Windlasses,	

Learning Outcome	Content	Suggested Assessment Methods
	– Functions and Uses of Capstans and Related Equipment • Signals For Operation Used As Per COSWP And ISM Code And Company Policy	
3. Undertake cargo handling gear and equipment operations	• Conventional Hatch Covers • Conventional Cargo Hold – Construction – Tank Top Ceiling – Spar Ceiling – Limber Boards – Bilge Suction- Strum Box – Deep Tanks • The Macgregor Steel Hatch Cover • General Cargo Terminology – Bale Space – Broken Stowage – Deadweight Cargo – Dunnage – Grain Space – Measurement Cargo – Stowage Factor • Hold Preparations • Stowage Methods: – Bagged Cargo – Barrels – Coal – Copra – Cotton – Edible Oils – Flour – Fruits – Glass – Vehicles • Cargo Handling: – Use of Snotters	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Use of The Bale Sling Strop – Palletization – Cargo Nets – Timber Dogs – Plate Clamps – Slings Sheet Metal – Use of Chain Slings • Ventilation – Natural Ventilation – Mechanical Ventilation • Refrigerated Cargoes – Typical Refrigerants – Properties of Good Insulating Materials – Loading Refrigerated Goods • Dangerous/Hazardous Cargo: – Procedure for Loading and Transportation of Hazardous Goods – Monitoring of Hazardous Cargoes • Cargo plans: – General cargo vessels – Cargo hold and hatch cover plan • Container Definitions and Terms – Administration – Approval /Approved – Cargo – Cell – Cell Guide – Container – Container Spreader Beam – Corner Fitting – Existing Container – Flexible Box Ship – Gantry Crane – Hatchless Holds – International Transport	

Learning Outcome	Content	Suggested Assessment Methods
	– Karrilift – Lashing Frame/ Lashing Platform – Maximum Operating Gross Weight – Maximum Permissible Payload – New Container – Owner – Prototype – Rating – Stack – Safety Approval Plate – Tare Weight – TEU – Type of Containers and Type-Series Container • Container Markings • Loading Containers • Container Tonnage • Loadicators and Loading Plan Computers • Tankers	
4. Rig and unrig bosun's chairs staging, pilot ladders, hoists, rat guards and gangways	• Introduction to Rope Work • Types of Ropes: – Natural Fibre Ropes – Manilla – Sisal – Coir – Hemp • Lay of Rope – Eight Strand Plaited – Shroud Lay – Soft-Laid – Hard-Laid – Standard or Plain Laid – Sennet Laid – Unkinkable Laid • Small Stuff	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Hambroline – Houseline – Boat Lacing – Marline – Spunyarn – Point Line – Ratline Logline – Lead Line – Steaming Twine – Roping Twine – Single Halyard • Synthetic Fibre Ropes – Nylon – Polyester – Polypropylene – Precautions When Handling Synthetic Man-Made Ropes • Bends and Hitches – Blackwall Hitch- Single – Blackwall Hitch- Double – Bowline – Bowline On the Bight – Running Bowline – French Bowline – Carrick Bend- Single – Carrick Bend- Double – Carrick Bend- (Single) Diamond – Catspaw – Clove Hitch – Cow Hitch – Figure Eight Knot – Fisherman’s Bend – Marline Spike Hitch – Midshipman’s Hitch – Reef Knot – Rolling Hitch – Round Turn and Two Half Hitches	

Learning Outcome	Content	Suggested Assessment Methods
	– Sheepshank – Sheetbend- Single – Sheetbend- Double – Timber Hitch – Barrel Slings • Working Aloft and Overside: – Rigging The Bosun’s Chair – Rigging Stages • Seizing: – Flat Seizing – Racking Seizing – Rope Seizing – Round Seizing • Ropework and Cordage Tools: – Hand Fid – Riggers’ (Sweden) Fid – Setting Fid – Serving Board – Serving Mallet • Worming Parcelling and Serving: – Worming – Parcelling • Cordage Splice: – Back Splice – Eye Splice – Short Splice – Long Splice • Whipping: – Common Whipping – Sailmakers Whipping – Palm and Needle Whipping – West Country Whipping • Marrying Two Ropes Together • To Pass a Stopper: – Common Rope Stopper – West Country (Chinese) Rope Stopper • Breaking Out Mooring Rope	

Learning Outcome	Content	Suggested Assessment Methods
	• Rigging and Unrigging of Gangway/MOT • Rigging and Unrigging Safety Nets • Inspection, Testing and Securing/Stowage of The Gangway/MOT • Monitoring And Reporting	
5. Operate anchoring equipment under various conditions	• Introduction • Types of Anchor: – Admiralty Pattern Anchor – The Stockless Anchor – Admiralty Cast Anchor – CQR (Plough Anchor) • Tests On Anchors: – Drop Test (Cast Anchors) – The Bending Test • Marks On Anchors • Anchor Certificate • Chain Cable Tests • Notes On Cable: – Accessories – Material of Manufacture – Size of Cable • Types of Shackles – Kenter Lugless Joining Shackle – ‘D’ Lugged Joining Shackle • Securing and Stowage of Anchors • Securing Anchor and Cable • Steam Windlass Operation • Preparing Anchor for ‘Letting Go’ • Brake Systems for Anchor Handling Windlass • Cable Holders • Procedure for Coming to Anchor: – Split Windlass Arrangement – Bell Signals – Marking of Anchor Cable • Clearing Away the Anchor (Past and Present Practice)	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Chain Cable/Stud Link General Information: – Volume of Anchor Chain – Anchor Chain Renewal – Size of Cable Measurement – Chain Grades – Grade Three Cable – Wrought Iron Cable – Baldt or Dilok Cable – Shackle Length – Strength • Anchor Terminology: – Anchor A-Cock Bill – Anchor Aweigh – Anchor Buoy – Anchor Coming Home – Anchor Dragging – Anchor Warp – Brought Up – Cable Clench – Cable Jack – Cross – Drop an Anchor Underfoot – Elbow – Foul Anchor – Foul Hawse – Grow – Gypsy – Hawse Pipes – Hove in Sight – Joggle Shackle – Kedging – Long Stay – Moored – Ream A Shackle – Render Cable – Round Turn – Scope – Shackle of Cable	

Learning Outcome	Content	Suggested Assessment Methods
	– Short Stay – Snub – Spurling Pipes – Tide Rode – Up and Down – Veer Cable – Walk Back Anchor – Wind Rode – Yaw • Watch at Anchor: – Transit Bearings – General Precautions – Vessel Dragging Towards Your Ship – Drawing Attention to The Situation – Anchor Party Forward – Use of VHF Radios • Anchoring Facility for High-Speed Crafts • Bruce (Mooring) Anchors • The ‘flipper’ delta anchor	
6. Use signals for the operations equipment	• Introduction • Standard Maritime communication (hand signals) • Communication methods: – Radio Communications – Public Address – Flag signaling terms – Single letter meanings (international code of signals) – Double letter meanings	• Observation • Written assessment • Oral assessment • Practical assessment
7. Implement operations completion procedure	• Health and safety at work precautions • Securing • Reporting	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
8. Operate access arrangements and pipeline systems	• Introduction • Bilge piping and pumping arrangement • Ballast piping and pumping arrangement	• Observation • Written assessment • Oral assessment • Practical assessment
9. Maintain mooring equipment	• Definition of terms • Mooring equipment: – Mooring lines - i.) Headline - ii.) Forward spring line - iii.) Breast line - iv.) Back spring line – Windlass – Mooring Winch – Bitts – Chocks – Bollards – Fairleads • Greasing • Rope/wire work • Mooring winch brake • Defects of mooring equipment reported as per company policy	• Observation • Written assessment • Oral assessment • Practical assessment
10. Maintain cargo handling gear and equipment	• Introduction • Lifting gear: – Derricks: – Single swinging derrick – Union purchase: - i.) Preventer guys - ii.) Slewing guys – Inspection and maintenance: - i.) visual inspection - ii.) Monthly inspection - iii.) Annual inspection - iv.) 5 yearly inspection • Maintenance of: – Hatch covers – Vent heads – Ventilation fans	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Ventilation • Hatch cover inspection and testing – High pressure test – Ultrasonic test • Reporting	
11. Maintain deck equipment	• Inspection: – Visual inspection – Monthly inspection – Annual inspection – 5 yearly inspection • Maintenance schedules • Reporting defects	• Observation • Written assessment • Oral assessment • Practical assessment

Suggested Methods Instruction

- Workshops
- Computer -Based Training (CBT) modules
- Shipboard training
- Demonstration equipment
- Simulators Project/ Lab
- Role play

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: - Windows/Linux/Macintosh Operating System - Microsoft Office Software - Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Fully equipped seamanship workshop • Snotters • Bale Sling Strop • Palletization • Cargo Nets • Timber Dogs • Plate Clamps • Chain Slings • Cargo loading software • Deck Machinery	• Seamanship Techniques by D.J House • COSWP Manual • Sample Company ISM Manual • 3 sets of synthetic Ropes of various sizes for each trainee • Steel wire Ropes of various sizes for each trainee • Seizing twine and seizing wire • 4 Heaving Lines • 8 Safety Belt

		• 8 Safety Harness • Twist locks • Turnbuckles • Lashing rods • Wires/ ropes • Photos of basic hand signals for machinery operations • Photos of anchors and Shackles • Photos of cargo purchases
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BERTHING, ANCHORING AND OTHER MOORING OPERATIONS

UNIT CODE: 1041 451 17A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform berthing anchoring and other mooring operations

Duration: 70 Hours

Unit Description

This unit specifies competencies required by Able Seafarer deck to contribute to berthing, anchoring and other mooring operations. It involves operating mooring equipment, operating anchoring equipment, handling mooring lines and tug lines, executing making Fast and letting go Procedures, implement anchoring operations, perform events associated with mooring to a buoy.

Summary of Learning Outcomes

1. Operate mooring equipment
2. Operate anchoring equipment
3. Handle mooring lines and tug lines
4. Execute making Fast and letting go Procedures
5. Implement anchoring operations
6. Perform events associated with mooring to a buoy

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Operate mooring equipment	• Introduction to Mooring operations • Meaning of terms – Mooring operations – Mooring equipment – Safe Working Load (SWL) – Working Load Limit (WLL) – Minimum Breaking Strength (MBS) – Single Up – Let go all – Clear from the pier – All fast – First Line Ashore – All Lines Onboard	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Occupational Health & safety precautions – Checklists – Work permits – Risk assessments – Tool box meetings – PPE – Tool selection • Mooring equipment – Capstans – Leads – Chocks – Hawses – Bollards – Winch – bitts • Mooring station General Arrangement • Communication media – VHF radios – Talk back system – Standard Hand signals • Types of mooring lines a. Nylon b. Polyester c. Polypropylene • Mooring lines system – Headlines – Stern lines – Spring lines – Breast lines – Messenger lines – Heaving lines – Towing lines – Stoppers – Rat- guards • Mooring winch operation – Heaving in – Slack out	

Learning Outcome	Content	Suggested Assessment Methods
	- Safe practices in handling mooring equipment - Testing - Maintenance - Operation	
2. Operate anchoring equipment	- Introduction to anchoring operations - Meaning of anchor terms - Heave up - Let go - Walk back - Brought up - Foul anchor - Long stay - Medium stay - Short stay - Up and down - Types of anchors - Stockless - Stock - Components of Anchor system - Anchors - Anchor lashing - Spurling Pipe - Chain locker - Chain stopper - Hawse pipe - Anchor shackle - Windlass gears and clutches - Turnbuckles - Pelican hook - Gypsy - Anchor chain - Markings - Common link - Kenter Link - Anchor winch operation - Heave in	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Slack out • Securing anchor for sea and anchorage	
3. Handle mooring lines and taglines	• Introduction to mooring line and tug lines • Safe practices in handling mooring and tug lines – Inspection – Preparation – Teamwork – Tension monitoring – Lead, Bitts and angles – Winch operation – Storage – Snap back zones – Emergency preparedness – Party to party coordination	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment
4. Execute make Fast and Let go Procedures	• Introduction to mooring procedures • Meaning of terms – Making fast – Let go • Making fast procedure – Preparation – Line handling – Making fast • Let go procedure – Preparation – Release sequence – Line retrieval	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment
5. Implement anchoring operations	• Inspection and readiness • Dropping anchor • Monitoring anchor • Retrieving anchor	• Observation • Portfolio of Evidence • Written assessment

Learning Outcome	Content	Suggested Assessment Methods
		• Practical assessment • Oral assessment
6. Perform functions associated with mooring to a buoy	• Introduction to mooring to a buoy • Meaning of terms – Single point mooring (SPM) – Bow line – pick-up line • Mooring to a buoy procedure – Inspection and readiness – Mooring to a buoy – Monitoring • Let go procedure – Release from buoy – Line retrieval	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project
- Role play
- Case study

Recommended Resources for 24 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard	• Fully equipped seamanship workshop • Steering console/navigation simulator	• Seamanship Techniques by D.J House • Bridge Procedures Guide by International Chamber of Commerce • COSWP Manual • Sample Company ISM Manual

• 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Manila Ropes [various sizes] • 3 sets of synthetic Ropes of various sizes for each trainee • Steel wire Ropes of various sizes for each trainee • Seizing twine and seizing wire • 4 Heaving Lines • 8 Safety Belt • 8 Safety Harness • 8 Marline Spikes • 8 Wooden Spikes • Access to Mooring Winch, hydraulic and pneumatic
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CARGO HANDLING AND STOWAGE

ISCED UNIT CODE: 1041 451 18A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform cargo handling and stowage

Duration: 80 hours

Unit description

This unit covers the competencies required by able seafarer deck to perform basic cargo handling and stowage. It involves contributing to cargo loading and unloading, contributing to cargo stowage and securing, contributing to cargo care during voyage and contributing to handling dangerous, hazardous and harmful cargo.

Summary of learning outcomes

1. Execute occupational health and safety procedures related to cargo handling.
2. Contribute to cargo stowage and securing.
3. Contribute to Cargo Loading and unloading.
4. Contribute to care for cargo during the voyage.
5. Contribute to handling dangerous, hazardous, and harmful cargo.
6. Recognize cargo documentation.

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Execute occupational health and safety procedures related to cargo handling.	• Terms used in cargo handling • PPE • Safe working procedures for cargo handling • Visual inspection of all cargo gear • Cargo gear test certificates and registration • Safe working load • Certificate of properties for ropes and wires • Inspection requirements • Working with hatch covers • Potentially dangerous spaces • Procedures for entering enclosed spaces • Safe working practices	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
2. Contribute to cargo stowage and securing	● Meaning of terms – cargo hold – Ro-Ro cargo – Stowage factor – Cargo handling – Loading – Unloading – Cargo – Dunnage – Broken stowage ● Types of cargo – Containerized cargo - i.) Machinery parts - ii.) Loose metals - iii.) Drums - iv.) Boxes – Non-Containerized cargo - i.) Tobacco - ii.) Rubber - iii.) Cement - iv.) fertilizers – Liquid Bulk cargo - i.) Crude oil - ii.) Fuel oils – Dry Bulk cargo: - i.) Grain - ii.) Iron - iii.) Coal - iv.) Sugar – Refrigerated cargo: - i.) fruits - ii.) dairy products - iii.) horticulture - iv.) meat – Dangerous. hazardous and harmful cargo: - i.) explosives - ii.) flammable gases - iii.) flammable liquids - iv.) flammable solids - v.) oxidizing substance and	● Observation ● Portfolio of Evidence ● Project ● Written assessment ● Practical assessment ● Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	organic peroxide vi.) toxic and infectious substances vii.) radioactive materials viii.) corrosives ● Meaning of Cargo holds/ tanks ● Preparation of Cargo holds/ tanks ● Cargo handling equipment – Cranes – Derricks – Slings – Slotters – Canvas slings – Trays – Pallets – Nets – Bale hooks – Chain slings – Vehicle sling ● Cargo loading/ discharge plan ● Regulations, procedures and instructions ● Loading/ unloading rate ● Cargo defects/ deviations ● Protection of sensitive equipment – Vent heads – Fans – Electrical components – Antennas – Wincos ● Final draught – Fore draught – Mid draught – Aft draught	
3. Contribute to cargo stowage and securing	● cargo securing equipment – Twist locks – Turnbuckles – Lashing rods – Wires/ ropes	● Observation ● Portfolio of Evidence ● Written assessment ● Practical

Learning Outcome	Content	Suggested Assessment Methods
	- adequate antidotes/ medicines related to cargo hazards/ properties - Precautions for handling cargoes - Safe access to cargo areas - Cargo stowage and securing - deviations from cargo stowage and securing plan - Cargo completion	assessment Oral assessment
4. Contribute to care for cargo during the voyage.	- Calibration of equipment - Shutting/opening of Ventilators - Cargo lashings - Sampling Cargo hold atmosphere - Monitoring Weather conditions - Precaution to the prevention of pollution of the marine environment - Reporting Leaks, vapors and uncharacteristic Odors - bilge soundings - Testing PH of Bilge water	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
5. Contribute to handling dangerous, hazardous and harmful cargo	- Meaning of terms - Classification of dangerous, hazardous and harmful cargo - Markings and labels - Occupational Health and safety precautions - Emergency response procedures in emergency situations - inspections of dangerous cargo - Reporting Leaks, spills or damage	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
6. Recognize cargo documentation.	- Cargo manifest - Cargo stowage Plan - Material Safety Data Sheet (MSDS) - Bill of Lading - Shippers' information	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment

Suggested methods instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: d. Windows/Linux/Macintosh Operating System e. Microsoft Office Software f. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Fully equipped seamanship workshop • Cargo handling gears	• COSWP Manual • Sample Company ISM Manual • 5 Cargo Securing manuals • 8 Safety Belt • 8 Safety Harness • 24 Helmet • 24 shoes • 24 pair of cargo handling Gloves • 24 Coveralls • 24 googles • Register of ships lifting appliances and cargo handling gear • 3 Code of safe working practice (COSWP) • 3 NLS cargo record book

SHIP HANDLING AND MANEUVERING

UNIT CODE: 1041 451 19A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform ship handling and maneuvering

Duration of Unit: 70 Hours

Unit Description

This unit covers the competencies required by an Able Seafarer Deck to contribute to ship handling and maneuvers. It involves under the directions of the Officer on Watch and the Master, performing ship handling in normal weather conditions and in adverse weather conditions, performing safe anchoring maneuvers, performing mooring operations, performing ship handling in emergencies and performing ship handling in shallow waters.

Summary of learning outcomes

1. Handle ship in normal weather conditions
2. Handle ship in adverse weather conditions
3. Perform safe anchoring manoeuvres
4. Perform mooring operations
5. Manoeuvre the ship in emergencies
6. Handle ship in shallow waters and narrow channels

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Handle ship in normal weather conditions	• Introduction • Terms and General Definitions – Cavitation – Con – Minimum steerage way – Drag – Headreach – Helmsman – List – Pitch – Pivot point – Set – Sheer – Squat – Stopping distance – Way	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Yaw • Steering and sailing rules (COLREGS) • Factors in Ship-Handling: a. Controllable Factors – Main engine power – Propeller(s). fixed or controllable pitch – Anchors – Mooring ropes – Rudder movement – Bow thruster – Bow rudder – Tugs b. Uncontrollable Factors – The weather – Tide and/or current – Geographical features (shallow water, floating obstructions, bridges, ice accretion) • High Speed Crafts and Safe Speed – Category ‘A’ craft – Category ‘B’ craft • Rudders: – Unbalanced Rudder – Semi-Balanced Rudder – Balanced Bolt Axle Rudder – Balanced Spade Type Rudder – The Schilling Rudder • Propellers: – Right-Hand Fixed Propeller – Controllable Pitch Propeller – Twin screw single rudder vessels – Pod propulsion units • Turning circles: – General definitions – Factors affecting turn • Steering applications: – Follow-up mode – Non-follow up mode	

Learning Outcome	Content	Suggested Assessment Methods
	– Remote mode – Automatic mode – Steering test applications – Short round turn	
2. Handle ship in adverse weather conditions	• Introduction • Terms and General Definitions – Heavy Weather – Adverse Weather • Handling of ship in Heavy and Adverse Weather: – Wind effects on ship handling – Effects of current on the motion of a ship – Use of current strength when turning in a channel – Simple turning and anchoring maneuvers • Handling ship in Tropical Revolving Storms (TRS) • Handling of ship during helicopter operations	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment
3. Perform safe anchoring manoeuvres	• Introduction • General anchor terms and definitions • The anchor plan • Anchor procedure: – Single anchor – Deep water anchoring – Laying/carrying out anchors – Clearing a foul hawse – To weigh anchor by deck tackle – Anchor recovery- Loss of windlass power – Hanging off an anchor – Lost anchor and re-securing of spare anchor • Anchor maneuvers: – Running moor	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Standing moor – Open moor – Baltic moor – Mediterranean moor – Dredging down – Snubbing round • Emergency anchoring	
4. Perform mooring operations	• Introduction • Berthing procedures: – wind onshore, tidal conditions calm/with tide – wind off-shore, tidal conditions calm/with tide – berthing between two ships – clearing a berth, wind and tide astern – clearing a berth, wind and tide ahead – clearing a berth, no tugs, right handed fixed pitch propeller – clearing a berth, portside to, no wind or tide – clearing a berth, starboard side to, no wind or tide – securing to buoys, no wind or tide – securing to buoys, wind and tide ahead – securing to buoys, wind and tide astern – Letting go from buoys, wind and tide astern – Letting go from buoys, wind and tide ahead – Entering a dry dock, wind and tide astern • Rigging slip wires	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment
5. Manoeuvre the ship in emergencies	• Introduction • Emergency contingency planning • IAMSAR VOL III	• Observation • Portfolio of Evidence

Learning Outcome	Content	Suggested Assessment Methods
	• Man Overboard Maneuvers: – Initial actions – Anderson turn – Williamson turn – Scharnow turn – Double turn – Delayed turn – When MOB is not located – Rescue and recovery of survivors • Helicopter operations: – Landing – evacuating personnel	• Project • Written assessment • Practical assessment • Oral assessment
6. Handle ship in shallow waters and narrow channels	• Introduction • General terms & definitions – Squat – “Smelling the bottom” – Canal blockage – Bank effect – Cushion effect – Canal speed • Shallow water effect and squat – Interaction – Interaction in narrow channels – Recommended passing positions of vessels in opposition in a narrow channel – Interaction between large vessels and tugs – Bank and cushion effect – Factors influencing squat - i.) Speed of the vessel - ii.) Rpm in relation to canal speed - iii.) Type of bow construction a. Working with Tugs – Safe handling of tow lines – Girting/Girding – Long distance towing – Alternative towing methods	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– Composite tow line – Use of two tugs	

Suggested methods instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Simulation
- Case study

Recommended resources for 25 trainees.

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: a. windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted white board markers	• Fully equipped seamanship workshop • Access to a bridge simulator • Aldis lamp • Ships whistle • Bell and Gong • Lights and shapes • Compass • Radar and ARPA • Nautical Charts/ECDIS	• Convention on the International Regulations for Preventing Collisions at Sea, 1972 • Company ISM manual • Bell book • Anchor log • Deck logbook • Contingency Plans • Bridge procedures guide • Maneuvering booklet • Wheelhouse poster

RESPONSE TO EMERGENCY AT SEA AND IN PORT

UNIT CODE: 1041 451 20A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Respond to emergencies at sea and in port.

Duration: 60 Hours

Unit Description

This unit covers the competencies required by an Able seafarer to contribute to response to emergencies at sea and in port. It involves, identifying emergencies at sea and in port, maintaining safety of passengers and cargo in emergencies onboard, performing damage control, performing actions taken following collision and grounding, executing emergency response procedures in Port, perform rescue operations and emergency response to distress at sea.

Summary of Learning Outcomes

1. Identify emergencies at sea and in port.
2. Maintain Safety of Passengers and Cargo in emergencies onboard
3. Perform damage control
4. Perform actions taken following collision and grounding
5. Execute emergency response procedures in Port
6. Perform rescue operations and emergency response to distress at sea

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Identify emergencies at sea and in port.	• Types of emergencies – Conditions. – Stranding/ Abandon ship – Allision – Cargo/passenger emergencies – Collision – Emergencies at anchor – Emergencies in port – Equipment failures – Main engine – Failure of critical navigational aid equipment on the bridge – Failure of the vessel's	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	alarm systems – Failure of the vessel's autopilot and steering systems – Failure of the vessel's navigation lights – Failure of the vessel's propulsion systems – Failures of bridge control equipment – Fires and Explosions. – Flooding and stability emergencies – Hazardous Material Spills. – Incidents harmful to the marine environment – Loss of stability – Main Engine failure – Man Overboard. – Maritime Terrorism and Piracy – Medical Emergencies. – Severe Weather Groundings • Master information of emergency • Types of alarms – General Alarm: – Fire Alarm – Man Overboard Alarm – Abandon Ship Alarm – Navigational Alarm – Machinery Space Alarm – Machinery Space CO2 Alarm – Cargo Space CO2 Alarm	

Learning Outcome	Content	Suggested Assessment Methods
2. Maintain Safety of Passengers and Cargo in emergencies onboard	● Immediate response actions ● Safety and security procedures ● Personal protective equipment (PPE) – Safety gloves – Helmet – Googles – Coverall ● Emergency equipment – Lifeboats – Lifejacket – Life raft – Lifebuoys – SART – EPIRB – Portable fire extinguisher – Fixed fire extinguisher ● Emergency Orders ● Allocated duties for emergency situations – Immediate assessments – Shore side response initiation – Substantial emergency response action – Additional reporting – Follow-up actions – Evolution to post-emergency conditions ● Emergency response Communications – Abandon ship signal – Man, overboard signals – Fire alarm – General emergency alarm	● Observation ● Portfolio of Evidence ● Project ● Written assessment ● Practical assessment ● Oral assessment
3. Perform damage control	● damage identification ● damage assessment ● actions to contain damage ● Factors creating or increasing risk of injury or damage	● Observation ● Portfolio of Evidence ● Project ● Written assessment

Learning Outcome	Content	Suggested Assessment Methods
	- Measures taken to relieve an emergency situation - Changes in conditions and behavior	- Practical assessment - Oral assessment
4. Perform actions taken following collision and grounding	- risk assessment procedures - tanks sounding monitoring - depth and nature of bottom ascertaining - Factors create or increase risk of injury or damage - Measures taken to relieve an emergency situation - conditions and behavior changes	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
5. Execute emergency response procedures in Port	- Master information of emergency - Alarms - preliminary actions - situation assessment - resources and equipment mobilization - emergency handling - emergency responses - documentation of events - report preparation	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
6. Perform rescue operations and emergency response to distress at sea	- Components of Distress messages - Identification, position, nature and kind of assistance required - Distress signals - Visual signals, such as a flame, a red flare, an orange smoke signal, or a square flag displayed with a ball - Automated radio signals, such as a Search and Rescue Transponder (SART) or an Emergency Position-Indicating Radio	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	Beacon (EPIRB) – Shapes and daylight signals, such as a black flag or a black ball – Inverted flags or ensigns – Visual manual HELP signal exhibited by the action of Slowly and repeatedly raising and lowering the arms outstretched to each side – Audio-Visual signaling by Using sound signaling equipment for SOS • Nature of assistance required • relay emergency • response to emergency • conducting a search – Proceeding to the area of distress – On board preparation – Action by assisting ships – Approaching the scene, radar search – Arrival on scene, implement search plan – Assistance by Search and Rescue (SAR) aircraft – Homing on radio signals – Aircraft casualties • Communications are maintained • Rescue operations	

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions

- Project
- Role play

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: a. windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted white board markers	• Fully equipped seamanship workshop • Access to a bridge simulator • Aldis lamp • Ships whistle • Bell and Gong • Lights and shapes • Compass • Radar and ARPA • Nautical Charts/ECDIS • 2 Survival Crafts • 3 GMDSS survival craft VHF radios • 2 EPIRBs • 2 SARTs • 1 CO2 fixed system • Foam installation including semi-portable and fixed system • Sprinkler system • Fire pumps (main and emergency fire pump) • Fire hoses, Hydrants, Branches and international shore connection • Decision making support system (DSS)	• 5 Company ISM manual • 5 Emergency plan • 5 IAMSAR manual • 5 National emergency response plans • GMDSS Equipment • 25 Lifejackets • 4 Lifebuoys • 5 Immersion suits and other thermal protective aids • 5 Rocket line throwing appliances • 5 Pyrotechnic (distress signals) • 25 PPE • 25 Whistles • Solid Materials (wood, textiles, curtains, furniture, plastics) • Liquid Materials (Patrols, Oils, Lubricants, paints, waxes, alcohol) • Portable Fire Extinguishers (foam, water, CO2, dry chemical)

General Resources	Tools and Equipment	Materials and Supplies
		• 4 Fire blankets • 4 Damage control plan • 4 Towing booklet • 4 IMDG code • 4 IMO/WHO/ILO Medical first aid guide for use in accidents involving dangerous goods (MFAG)



CHARTWORK

UNIT CODE: 1041 451 21A

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform basic chartwork practices

Duration: 80 Hours

Unit Description

This unit covers the competencies required by the Able Seafarer Deck to perform basic chartwork. It involves using earth features for navigation, using charts and nautical publications, using navigation aids, reading courses and bearings on compasses, plotting dead reckoning DR position and identifying the IALA buoyage system.

Summary of Learning Outcomes

1. Use earth features for navigation
2. Use charts and nautical publications
3. Use navigation aids
4. Read courses and bearings on compasses
5. Plot dead reckoning DR position
6. Identify the IALA buoyage system

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Use earth features for navigation.	● Shape of the earth and its dimensions ● Meaning of earth features ● Types of earth features ● Earth direction – True direction – True north – Navigational compass – True bearing – True course – True heading – Position by bearing and distance ● Datum of nautical charts ● Latitudes and longitudes ● Way points creation ● Measurement of distance	● Observation ● Written assessment ● Oral assessment ● Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	between two points on a Mercator chart Distance and time estimation	
2. Use charts and nautical publications	- Definition of charts and nautical publications - Properties of nautical charts - Types of charts - Natural scale of charts - Chart projections - Navigation area selection - Types of nautical publications - Lists of radio signals - Sailing directions - Tidal publications - Lists of lights and fog signals - Distance table. - Chart catalogue. - Mariners' handbook - Nautical almanac - Notice to mariners - Ocean passages of the world - IALA maritime buoyage system - Chart 5011 - Nautical chart and publication updates - Correction log	- Observation - Portfolio of Evidence - Written assessment - Practical assessment - Oral assessment
3. Use navigation aids	- Meaning of navigation aids - Types of navigation aids - Buoy - Fog signals - Light houses - Day beacons - Bearing with navigation aids - Position lines with transit bearing - IALA buoyage system markings	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Visual navigation – Daytime – Night • Ships course • Navigational aids log	
4. Read courses and bearings on compasses	• Definition of terms – Course – Bearing – True course – Magnetic course • Types of compasses – Magnetic compass – Gyrocompass • Basic parts of compasses – Magnetic compass – Gyro compass • Compass corrections – Deviation – Variation – Gyro error (high or low) – Compass error • Compass courses notation • Compass bearing – True bearing – Relative bearing – Magnetic bearing	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment
5. Plot dead reckoning DR position	• Definition of terms – Dead reckoning position (DR) – Position lines – Fix – Course to steer • DR position plotting • Known position plotting • Compass errors adjustments – deviations and variations from tables and charts – true course from compass course	• Observation • Portfolio of Evidence • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	– compass course from true course – compass error using a transit bearing – compass error to the ship's head and compass bearings to convert to true • Distance calculation • Course to steer plotting and marked.	
6. Identify IALA buoyage system	• Meaning and purpose of IALA buoyage system • Types of IALA Buoyage systems – Region A – Region B • IALA buoyage system publication – Admiralty NP735 • IALA buoyage system interpretation on charts • IALA buoyage system marks – Lateral marks – Cardinal marks – Isolated danger marks – Safe water marks – Special marks – Emergency wreck marking buoy	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – windows/Linux/Macintosh Operating System – Microsoft Office Software – Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted white board markers	• Compass • Radar and ARPA • Nautical Charts/ECDIS • Sextant • Azimuth mirror • Echo sounders • GPS/DGPS • Course recorder • AIS • Speed log (Doppler, electromagnetic log) • VDR/SVDR • GMDSS Equipment • Radar • ARPA • ECDIS • Magnetic compass • Gyro Compass • Binoculars	• Ships logbooks • Flags and shapes (international code of signals) • Navigational charts and publications: – 25 harbor charts – 25 general charts – 25 berthing charts – 25 Sailing charts – 25 approaches charts – 25 coastal charts – 6 Lists of radio signals publications – 6 Sailing directions publications – 6 Tidal publications – 6 Lists of lights and fog signals publications – 6 Distance table. Publications – 6 Chart catalogue. Publications – 6 Mariners' handbook publications – 6 Nautical almanacs – 6 Notice to mariners' publications – 6 Ocean passages of the world publications

General Resources	Tools and Equipment	Materials and Supplies
		– 6 IALA maritime buoyage system (Admiralty NP735) – 6 Chart 5011 ● 25 Parallel rulers, divider, pair of compasses, chart correction pen ● Company ISM manual



ANNEX 1 – STCW TABLE A-II/5

This table shall provide training and assessment guidance during the implementation of this curriculum.

Function: Navigation at the support level

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Contribute to a safe navigational watch	Ability to understand orders and to communicate with the officer of the watch on matters relevant to watchkeeping duties Procedures for the relief, maintenance and handover of a watch Information required to maintain a safe watch	Assessment of evidence obtained from in-service experience or practical test	Communications are clear and concise Maintenance, handover and relief of the watch is in conformity with acceptable practices and procedures
Contribute to berthing, anchoring and other mooring operations	Working knowledge of the mooring system and related procedures, including: 1. The function of mooring and tug lines and how each line functions as part of an overall system 2. The capacities, safe working loads, and breaking strengths of mooring equipment, including mooring wires, synthetic and	Assessment of evidence obtained from one or more of the following: 1. Approved in-service experience 2. Practical training 3. Examination 4. Approved training ship experience 5. Approved simulator training, where appropriate	Operations are carried out in accordance with established safety practices and equipment operating instructions

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	fibre lines, winches, anchor windlasses, capstans, bitts, chocks and bollards 3. The procedures and order of events for making fast and letting go mooring and tug lines and wires, including towing lines 4. The procedures and order of events for the use of anchors in various operations 5. Working knowledge of the procedures and order of events associated with mooring to a buoy or buoys		

EXCELLENCE IN MARITIME TRAINING & DEVELOPMENT

Function: Cargo handling and stowage at the support level

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Contribute to the handling of cargo and stores	Knowledge of procedures for safe handling, stowage and securing of cargoes and stores, including dangerous, hazardous and harmful substances and liquids Basic knowledge of and precautions to observe in connection with particular types of cargo and identification of IMDG labelling	Assessment of evidence obtained from one or more of the following: 1. Approved in-service experience 2. Practical training 3. Examination 4. Approved training ship experience 5. Approved simulator training, where appropriate	Cargo and stores operations are carried out in accordance with established safety procedures and equipment operating instructions The handling of dangerous, hazardous and harmful cargoes or stores complies with established safety practices



Function: Controlling the operation of the ship and care for persons on board at the support level

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Contribute to the safe operation of deck equipment and machinery	Knowledge of deck equipment, including: 1. Function and uses of valves and pumps, hoists, cranes, booms, and related equipment 2. Function and uses of winches, windlasses, capstans and related equipment 3. Hatches, watertight doors, ports, and related equipment	Assessment of evidence obtained from one or more of the following: 1. Approved in-service experience 2. Practical training 3. Examination 4. Approved training ship experience	Operations are carried out in accordance with established safety practices and equipment operating instructions
Contribute to the safe operation of deck equipment and machinery (continued)	4. Fibre and wire ropes, cables and chains, including their construction, use, markings, maintenance and proper stowage 5. Ability to use and understand basic signals for the operation of equipment, including winches,	Assessment of evidence obtained from practical demonstration	Communications within the operator's area of responsibility are consistently successful Equipment operation is safely carried out in

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	windlasses, cranes, and hoists 6. Ability to operate anchoring equipment under various conditions, such as anchoring, weighing anchor, securing for sea, and in emergencies Knowledge of the following procedures and ability to: 1. Rig and unrig bosun's chairs and staging 2. Rig and unrig pilot ladders, hoists, rat-guards and gangways 3. Use marlin spike seamanship skills, including the proper use of knots, splices and stoppers Use and handling of deck and cargo-handling gear and equipment:	Assessment of evidence obtained from practical demonstration Assessment of evidence obtained from practical demonstration	accordance with established procedures Demonstrate the proper methods for rigging and unrigging in accordance with safe industry practice Demonstrate the proper creation and use of knots, splices, stoppers, whippings, servings as well as proper canvas handling

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	- Access arrangements, hatches and hatch covers, ramps, side/bow/stern doors or elevators - Pipeline systems – bilge and ballast suctions and wells		
Contribute to the safe operation of deck equipment and machinery <i>(continued)</i>	Cranes, derricks, winches Knowledge of hoisting and dipping flags and the main single-flag signals. (A, B, G, H, O, P, Q)		Demonstrate the proper use of blocks and tackle Demonstrate the proper methods for handling lines, wires, cables and chains
Apply occupational health and safety precautions	Working knowledge of safe working practices and personal shipboard safety including: - working aloft - working over the side - working in enclosed spaces - permit to work systems - line handling - lifting techniques and methods	Assessment of evidence obtained from one or more of the following: - approved in-service experience - practical training - examination - approved training ship experience	Procedures designed to safeguard personnel and the ship are observed at all times Safe working practices are observed and appropriate safety and protective equipment is correctly used at all times

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	of preventing back injury 7. electrical safety 8. mechanical safety 9. chemical and biohazard safety 10. personal safety equipment		
Apply precautions and contribute to the prevention of pollution of the marine environment	Knowledge of the precautions to be taken to prevent pollution of the marine environment Knowledge of the use and operation of anti-pollution equipment Knowledge of the approved methods for disposal of marine pollutants	Assessment of evidence obtained from one or more of the following: 1. approved in-service experience 2. practical training 3. examination 4. approved training ship experience	Procedures designed to safeguard the marine environment are observed at all times
Operate survival craft and rescue boats	Knowledge of the operation of survival craft and rescue boats, their launching appliances and arrangements, and their equipment Knowledge of survival at sea techniques	Assessment of evidence obtained from approved training and experience as set out in section A-VI/2 , paragraphs 1 to 4	Actions in responding to abandon ship and survival situations are appropriate to the prevailing circumstances and conditions and comply with accepted safety practices and standards

Function: Maintenance and repair at the support level

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Contribute to shipboard maintenance and repair	Ability to use painting, lubrication and cleaning materials and equipment Ability to understand and execute routine maintenance and repair procedures Knowledge of surface preparation techniques Understanding manufacturer's safety guidelines and shipboard instructions Knowledge of safe disposal of waste materials Knowledge of the application, maintenance and use of hand and power tools	Assessment of evidence obtained from practical demonstration Assessment of evidence obtained from one or more of the following: 1. approved in-service experience 2. practical training 3. examination 4. approved training ship experience	Maintenance and repair activities are carried out in accordance with technical, safety and procedural specifications